Energy – Work conservation of energy

Physics / Edition 8
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Energy

- **Energy** is defined as the measure of a system's ability to do work. That is to apply a force over a distance. This idea of energy was unknown to Newton. If an object has energy, damage can be done to another Object by going into it = transfer of energy
  - We use the symbol E to represent energy.
  - Energy has the same units as work:
    - Joule for SI, ft·lb for English

UNITS again

- The units of energy:
  - Metric
    - SI: joule ($J = N \cdot m$),
    - erg ($= 10^{-7}$ J = 0.0000001),
    - calorie (cal = 4.186 J),
    - 1 food Cal = 1000 cal=4,186 J
    - kilowatt-hour (kWh = 0.278 J).
  - English:
    - foot-pound ($ft \cdot lb$),
    - British thermal unit (Btu).

FORMS OF ENERGY

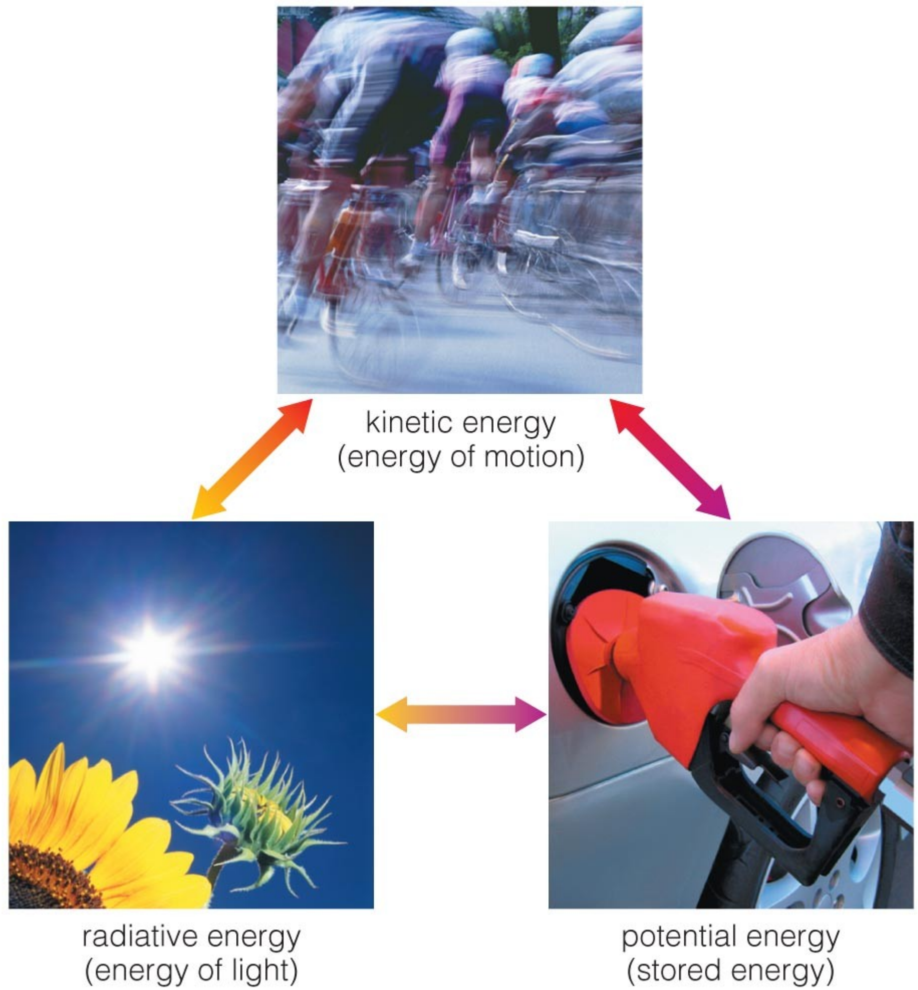
Kinetic (motion)

Radiative (light)

Potential (stored)

Energy can change type, but cannot be created or destroyed.

Energy can be converted from one form to another.



KINETIC ENERGY

- There are various types of energy.
  - **Kinetic energy** is the energy associated with an object's motion.
    - We use the symbol *KE*. $KE = \frac{1}{2} m V^2$
    -

Work= Change in kinetic energy : $\Delta KE = 0.5\ m\ (Vf^2 - Vi^2)$

Note that the kinetic energy depends on the speed squared

So if you double your speed on the highway , damage during a potential Crash is multiply by 4 !!! This was not known to Newton. Leibniz (co-founder of calculus) got the idea after an experiment was done with dropping balls in clay . Double the speed the balls have, Multiply the marks in the clay by 4 !!

Example:

1) A 50kg child runs at a speed of 5m/s and stop.
What is the change in kinetic energy ? HINT:

First find the Final kinetic energy
Then the initial kinetic energy then :

Change= FINAL – INITIAL

2) A person has a mass of 45kg and is moving with a velocity of 10m/s.
Find the person's kinetic energy?

The person's velocity becomes 5m/s. What is the kinetic energy ?

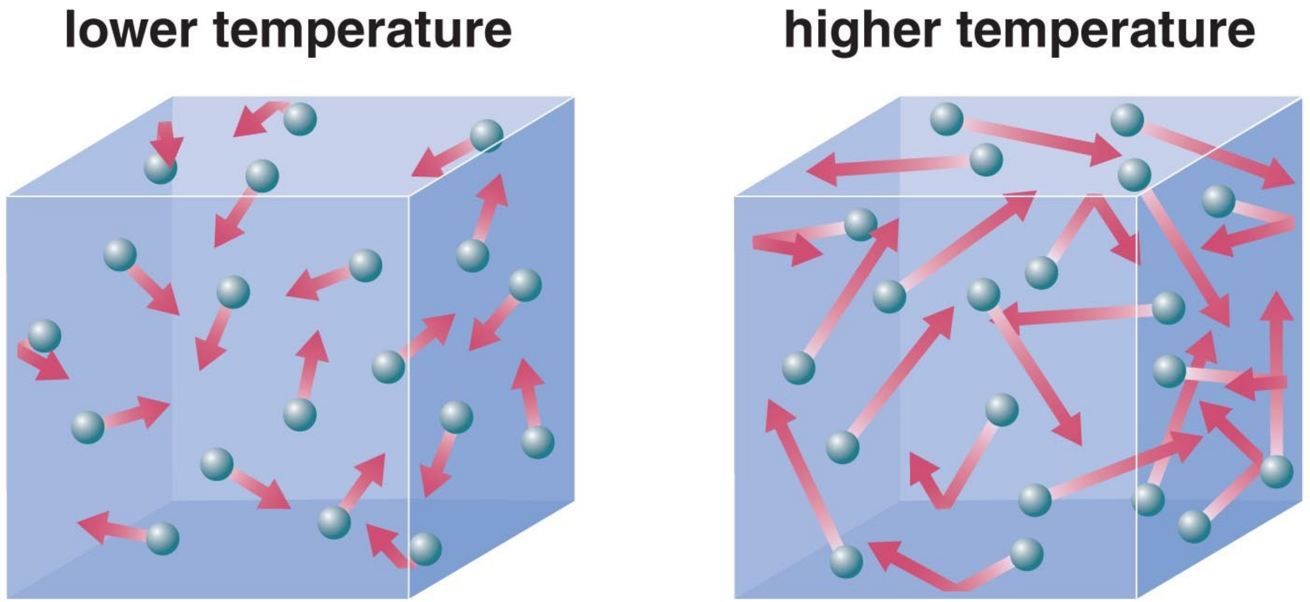
What is the ratio of the speeds ?
What is the ratio of the kinetic energy ?
Conclusion ?

Thermal Energy:

the collective kinetic energy of many particles (for example, in a rock, in air, in water)

Thermal energy is related to temperature but it is NOT the same.

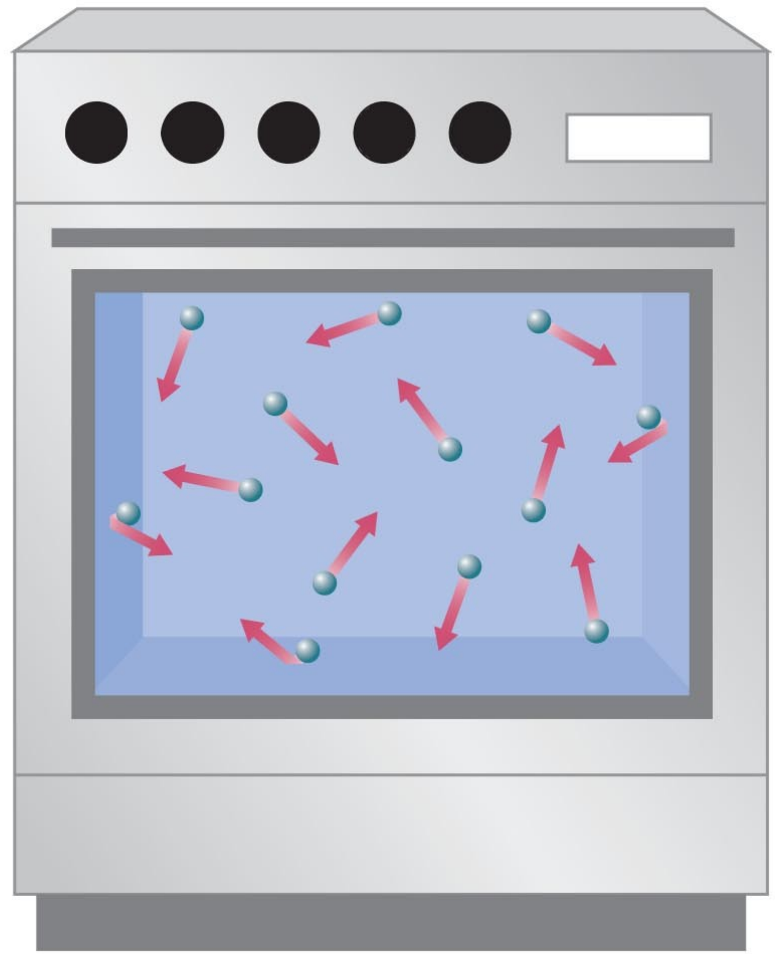
Temperature is the *average* kinetic energy of the many particles in a substance.



Thermal energy is a measure of the total kinetic energy of all the particles in a substance. It therefore depends on both *temperature* AND *density.*

Example:

212°F



400°F

POTENTIAL ENERGY IS STORED ENERGY. It's a binding energy. Or negative energy. Potential energy is energy associated with the system's position or orientation. We use the symbol PE.

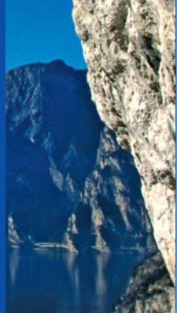
- CHEMICAL POTENTIAL ENERGY (plants, coal, oil, cookies ..)
The energy is in the bonds between atoms. These forces Are like compressed springs. The can expand suddenly. The forces are electromagnetic forces.

-NUCLEAR ENERGY = energy stores in the nuclei. $E = mc^2$
Energy stored in the bonds between nucleons.

- GRAVITATIONAL POTENTIAL ENERGY (a book above the ground)
Gravity bonds us to the Earth, bonds the stars in galaxies and galaxies in cluster. On Earth we compute the change in potential energy and we take PE = 0 at ground level.
$\Delta PE = m\, g\, \Delta h$ (g=10m/s/s)

- ELASTIC POTENTIAL ENERGY
(a compressed or stretched spring..) $\Delta PE = \frac{1}{2} K \left(\Delta x \right)^2$

Gravitational Potential Energy



$$W_{\text{gravity}} = mgh_o - mgh_f$$

DEFINITION OF GRAVITATIONAL POTENTIAL ENERGY

The gravitational potential energy PE is the energy that an object of mass *m* has by virtue of its position relative to the surface of the earth. That position is measured by the height *h* of the object relative to an arbitrary zero level:

$$\text{PE} = mgh$$

$$1\ \text{N} \cdot \text{m} = 1\ \text{joule}\ (\text{J})$$

Example

A 3-kg brick is lifted to a height of 0.5 meters above a table that is 1.0-m tall. Find the gravitational potential energy relative to the table and the floor.

$\Delta PE = m\ g\ x \Delta height$

Potential energy is relative to a reference level

You can take PE = 0 where ever it is convenient

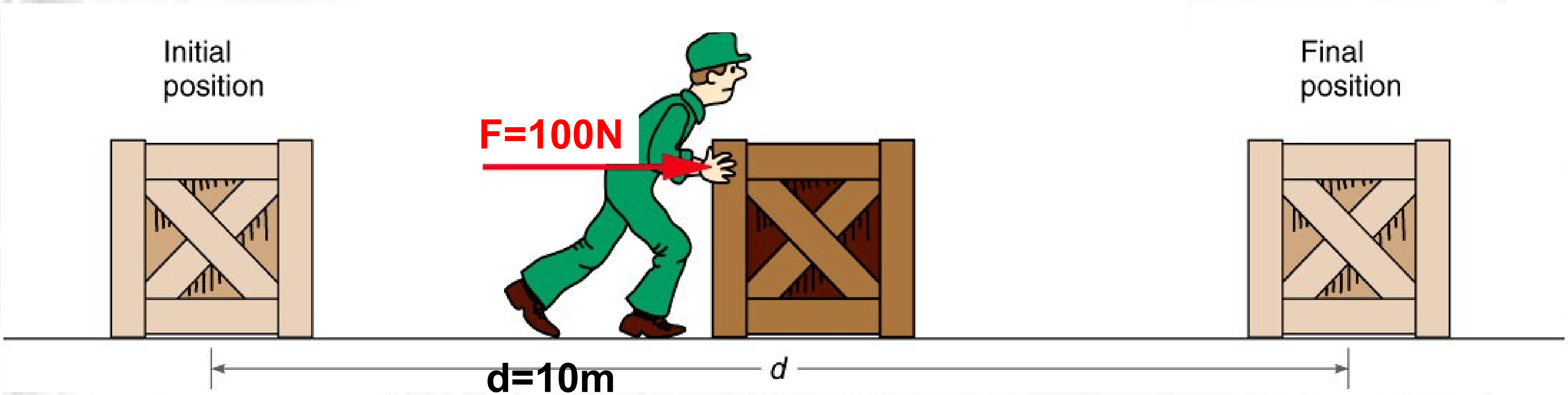
For computations. PE=0 at ground level.

The PE is meaningless without specifying the reference level.

In PHYSICS:
WORK MEASURES The AMOUNT of ENERGY of motion Transferred FROM one SYSTEM to another. Work on a system is done by a force only it changes the hinetic energy of the system.

Moving a mass long the x-axis - > work = Fx ΔX
Fx is the x-component of the force - ΔX is the displacement along x-axis

Work in joules, distance in meter, force F in newtons



Example: You push a box wth a force Of 100N and if the box is moving by 10m Then the energy transferred from your Body (energy stored in fat) to the box (energy of motion) = ________ x ___________ = ________ J

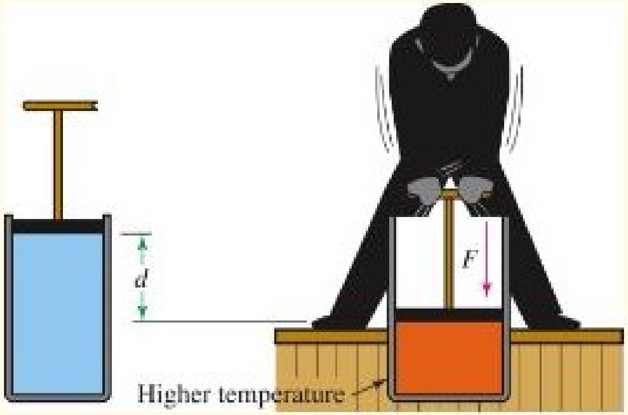
Work, cont'd

- Recall that force is a vector.
  - Involves magnitude and direction.
- Work is just that part of the force in the direction of the displacement.
  - Work is not a vector — it's a scalar.
- But the sign of the work does depend on the relative directions.

If work <0 energy is removed from the system

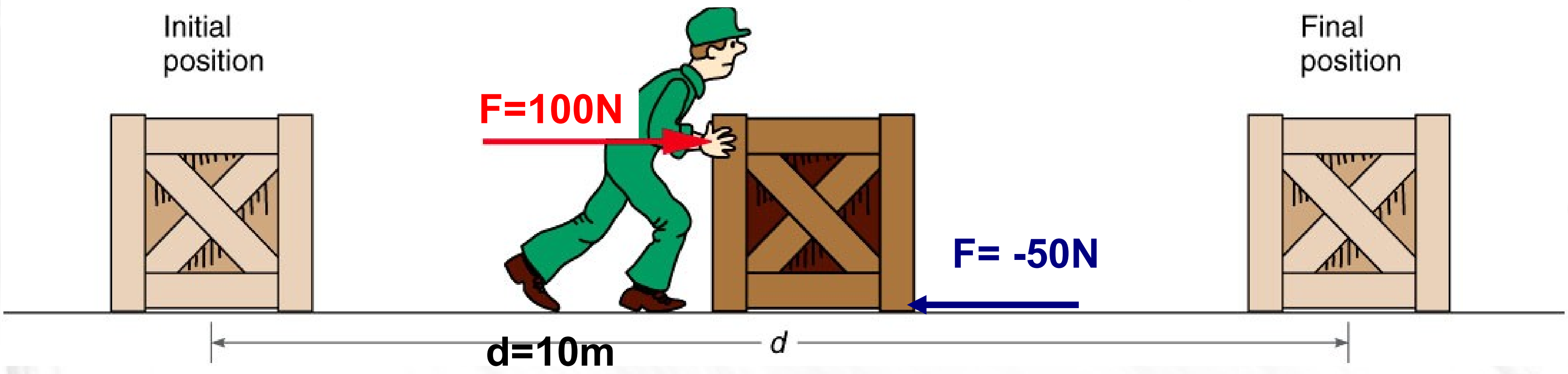
- **If work >0 energy is added to the system**
  also: work is done on a system only if the speed changes.

- Consider piston pushing on a gas in a cylinder.
  - Forcing the piston down requires a force.
  - Applying this force through a distance means you are doing work.
  - The work is done against the gas.
    - You compress the gas.
  - The gas gets hot.



- If the force and distance are in the same direction, the force does *positive work*.

If the force and distance are in the opposite direction, the force does *negative work*.



Work by the guy = ____ J work by the friction = ____ J
NET Work = ____J CHANGE IN ENERGY = ____________ J

A suitcase has a mass of 10kg.

Find its weight in newtons

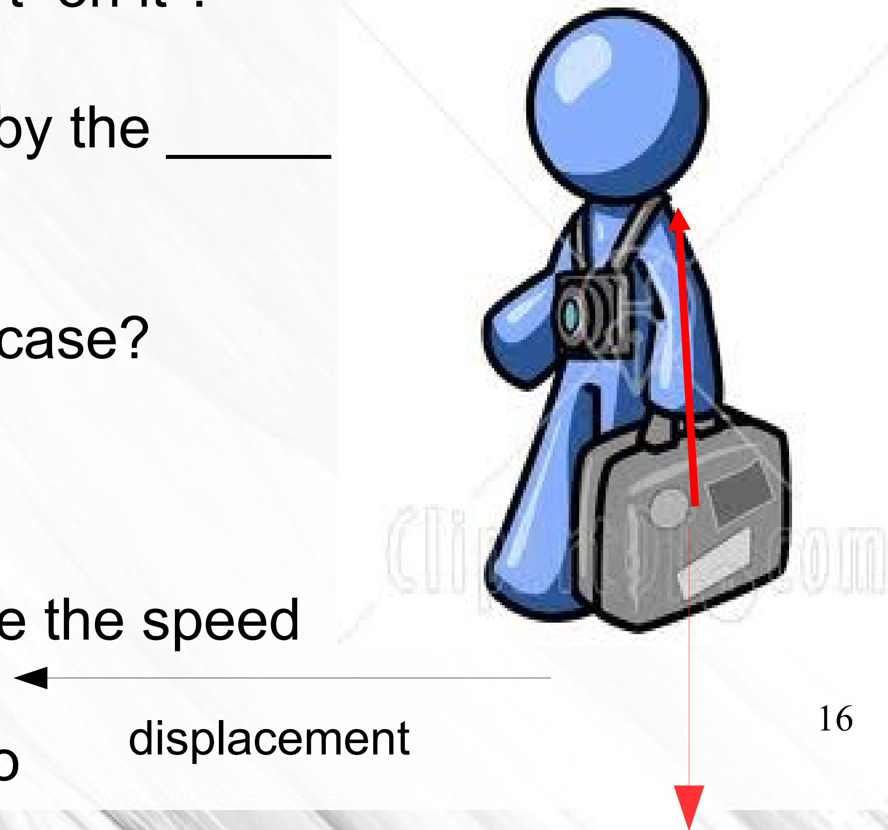
If you want to carry the suitcase , what is the
Upward force you need to exert on it ?

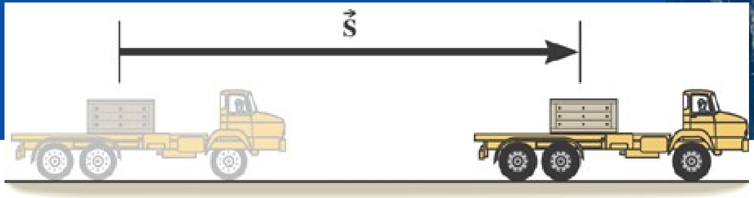
The upward force is balanced by the _____
of the suitcase.

Are you doing work on the suitcase?

Is gravity doing work ?

No because they don’t increase the speed
In the forward direction.
The forces are perpendicular to
Displacement. Wdone = 0



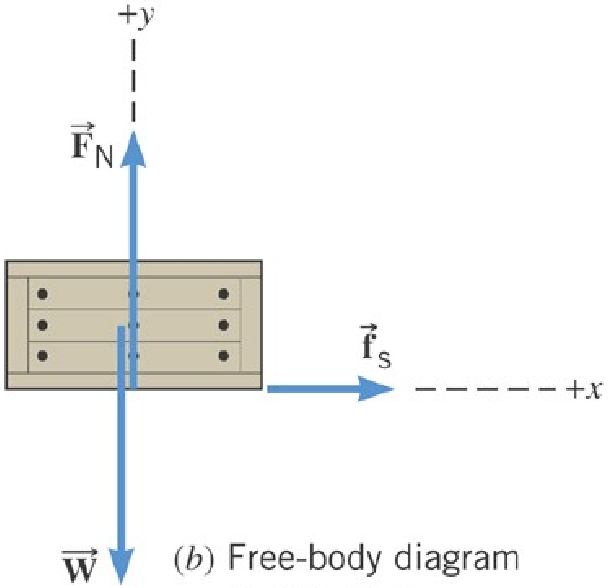


(a)

Example 3 Accelerating a Crate

The truck is accelerating at a rate of +1.50 m/s². The mass of the crate is 120-kg and it does not slip. The magnitude of the displacement is 65 m.

What is the total work done on the crate by all of the forces acting on it?



(b) Free-body diagram for the crate

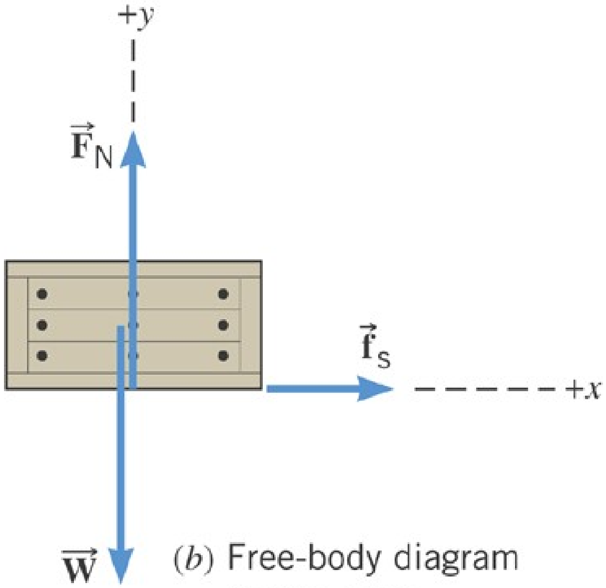
6.1 Work Done by a Constant Force

(a)

The angle between the displacement and the normal force is 90 degrees.

The angle between the displacement and the weight is also 90 degrees.

$$W = \left(F\cos 90\right)s = 0$$



(b) Free-body diagram for the crate

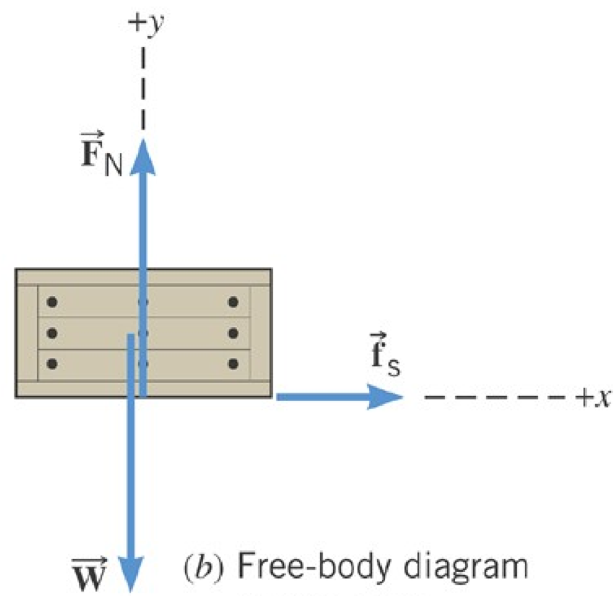
6.1 Work Done by a Constant Force

(a)

The angle between the displacement and the friction force is 0 degrees.

$$f_s = ma = (120\,\mathrm{kg})(1.5\,\mathrm{m/s^2}) = 180\mathrm{N}$$

$$W = [(180\mathrm{N})\cos 0](65\,\mathrm{m}) = 1.2\times10^4\,\mathrm{J}$$



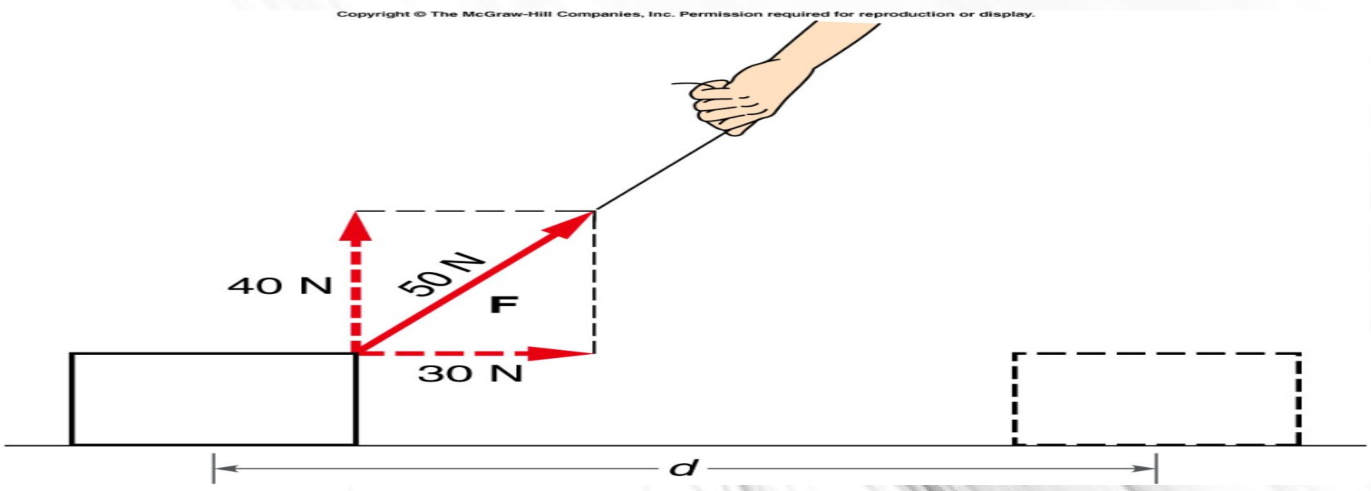
(b) Free-body diagram for the crate

If the force is not in the direction of the direction, the force does *no work*.

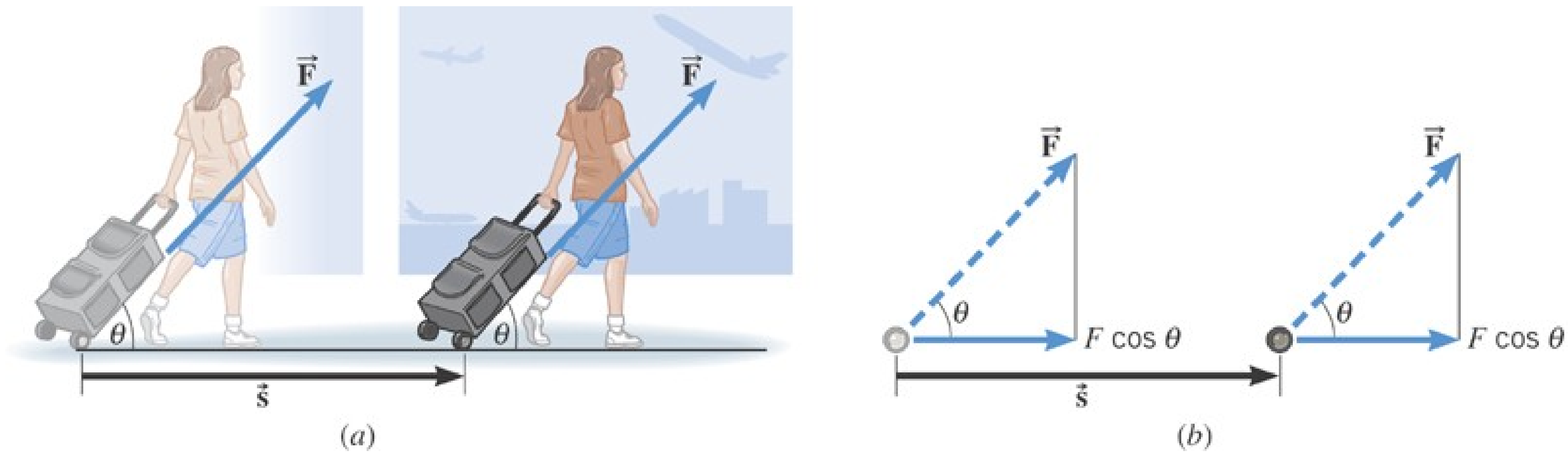
Bellow the pulling force for has 2 components, which component can increase the energy of the box (10kg) ?

So which component does work ?

Find the increase of energy ? (d=10m)



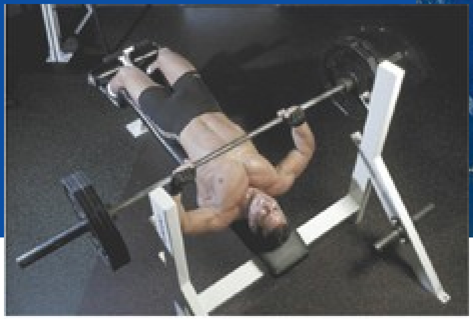

Total distance = 150 m
How much work does he do?
J
$\alpha = 60°$
$F_P = 20N$
m = 7 Kg



Example 1 Pulling a Suitcase-on-Wheels

Find the work done if the force is 45.0-N, the angle is 50.0 degrees, and the displacement is 75.0 m.

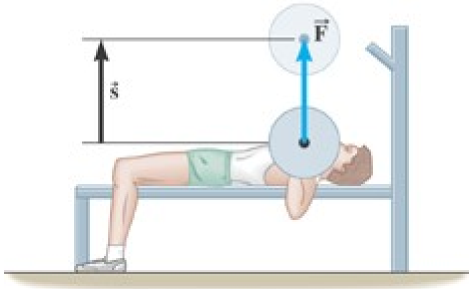
Textbook 7.1 and 7.3



(a)

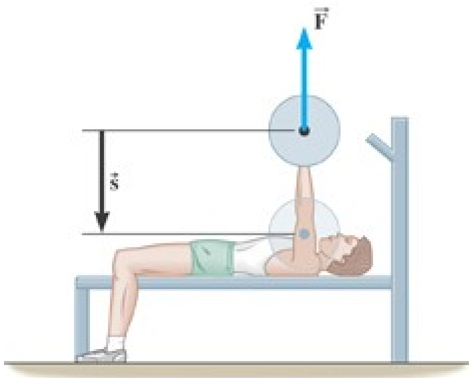
Lifting:
You are doing positive work (against gravity)
The displacement and the force have the same direction
Gravity is doing negative work (against displacement)
The total work is 0 (no change in speed for dumbbell)

$$W = (F\cos 0)s = Fs$$



(b)

$$W = (F\cos 180)s = -Fs$$



(c)

Down:
You are doing negative work (against displacement). Trying To slow the dumbbell down. You are still doing work against gravity. Gravity is doing positive work (same direction as displacement). Trying to speed up the dumbbell. Net work is 0

Gravity is a "conservative" force

A restoring force → tries to decrease the potential energy.
A spring force is also a restoring force.

DEFINITION OF A CONSERVATIVE FORCE

Version 1 A force is conservative when the work it does on a moving object is independent of the path between the object's initial and final positions.

Version 2 A force is conservative when it does no work on an object moving around a closed path, starting and finishing at the same point.

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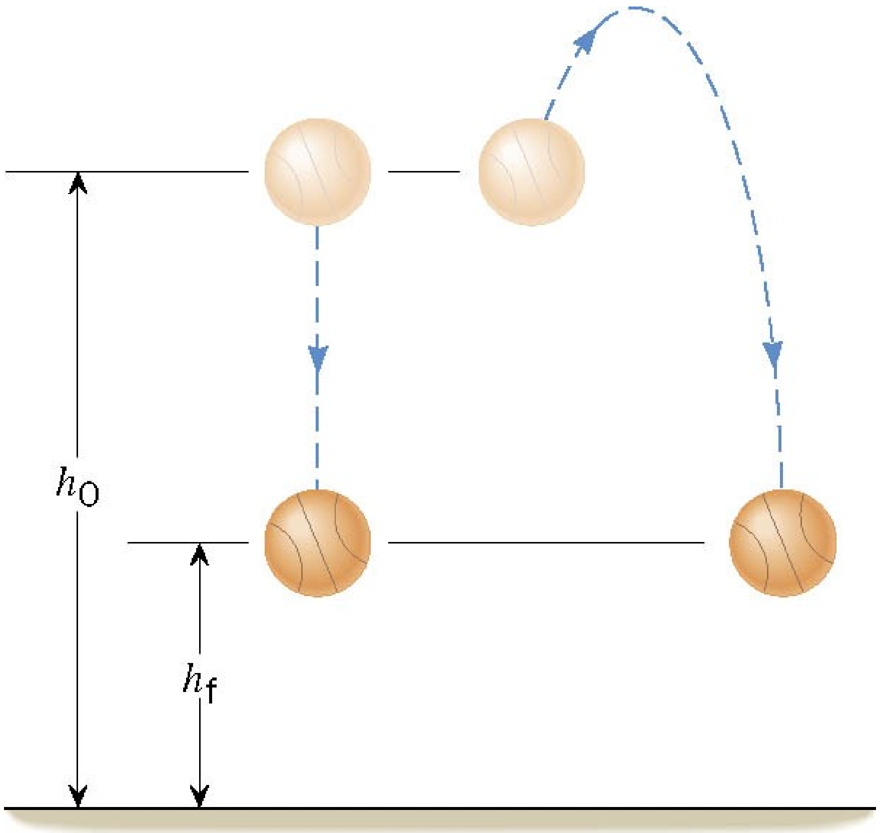
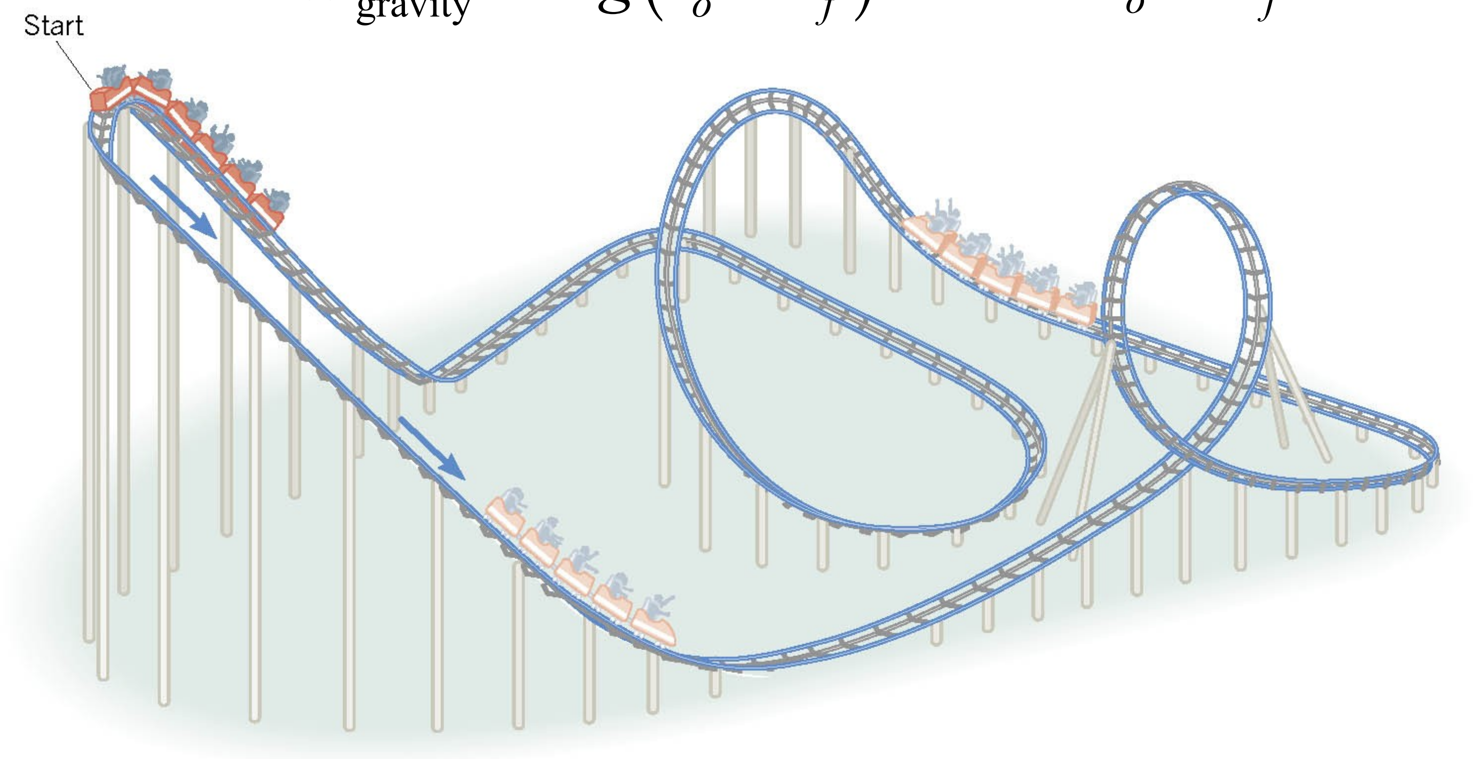
$$W_{\text{gravity}} = mg\left(h_o - h_f\right)$$



Version 2 A force is conservative when it does no work on an object moving around a closed path, starting and finishing at the same point.

$$W_{\text{gravity}} = mg\left(h_o - h_f\right) \qquad h_o = h_f$$



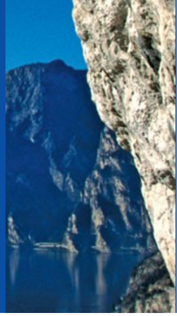


Table 6.2 Some Conservative and Nonconservative Forces

Conservative Forces	
	Gravitational force (Ch. 4)
	Elastic spring force (Ch. 10)
	Electric force (Ch. 18, 19)
Nonconservative Forces	
	Static and kinetic frictional forces
	Air resistance
	Tension
	Normal force
	Propulsion force of a rocket

6.4 Conservative Versus Nonconservative Forces

An example of a nonconservative force is the kinetic frictional force.

$$W = \left(F\cos\theta\right)s = f_k \cos 180^\circ s = -f_k s$$

The work done by the kinetic frictional force is always negative. Thus, it is impossible for the work it does on an object that moves around a closed path to be zero.

The concept of potential energy is not defined for a nonconservative force.

Work-Kinetic Energy Theorem

When work is done by a **net force** on an object and the only change in the object is its speed, the work done is equal to the change in the object's kinetic energy

Wnet = change in KE

- Speed will increase if work is positive
- Speed will decrease if work is negative

Example:

You are pushing a box with a force of 100N
But a 50N frictional force opposes the motion.

1) Find the work done by each force

2)Find the net work (distance is 5m)

3) Find the increase in kinetic energy

4) If the initial speed of the box is 0(starts from rest) and if the Box is 50 kg , find the final speed.
$KE=0.5\ m\ v^2$

Example

Because of friction, a constant force of 100 newtons is needed to slide a box across a room. If the box moves 3 meters, how much work must be done?

(that is how much work you are doing on the box).

If the friction is – 100N, what is the work done by the Friction force ?

What is the total net force ? the net work ?
The change in kinetic energy ?

1) A 3600kg boat is moving with a velocity of 2.4m/s. A much smaller 100kg jet ski has the same amount of kinetic energy. What is the velocity of the jet ski?

2) A7.00kg bowling ball travels a distance of 5.00m in 4.00s. How much kinetic energy does it have in J?
(hint: compute the speed first)

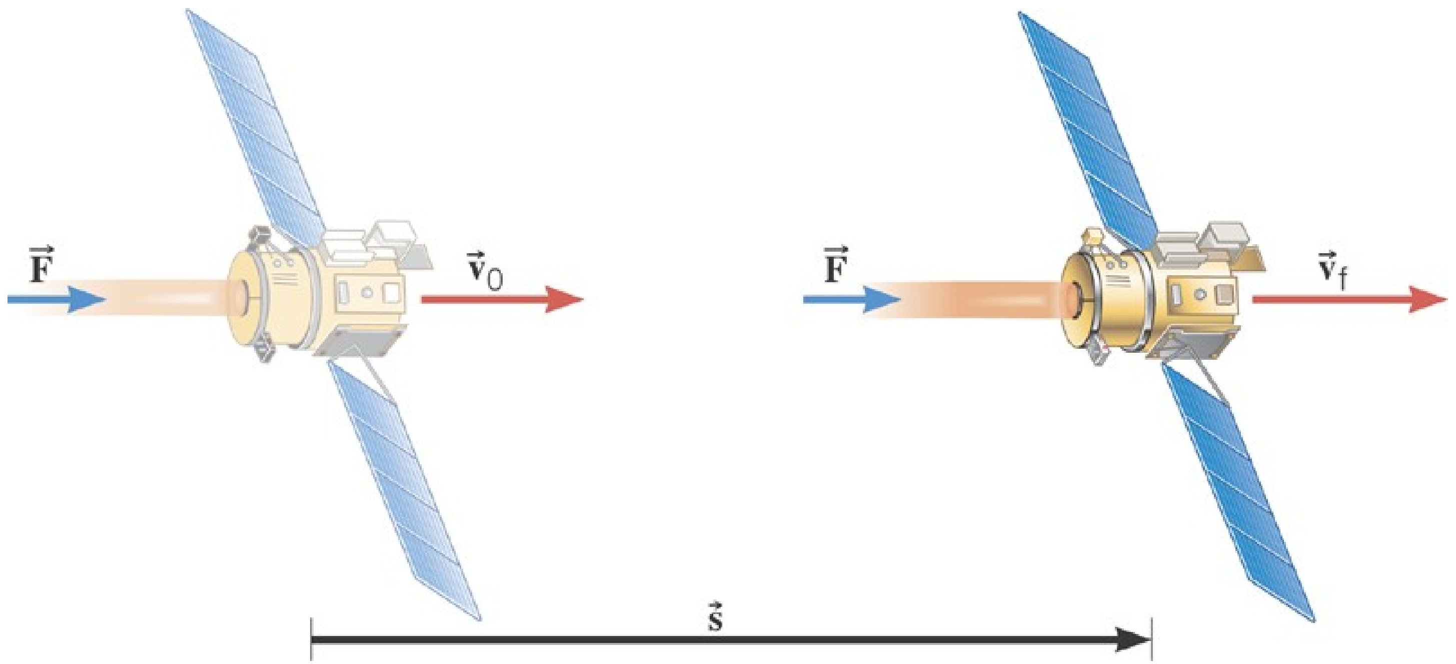
3) How much work does it take to make a 77.7kg wagon speed up from 0.8m/s to 3.3m/s?

4) How much work does a tugboat need to do in order to move a barge from rest to a velocity of 0.550m/s?
The mass of the barge is 879,000kg. The mass of the tugboat is insignificant.

5) How much work is done by gravity when a 7.64kg boulder falls to the ground from the top of a 33.4m tall cliff?

Example 4 Deep Space 1

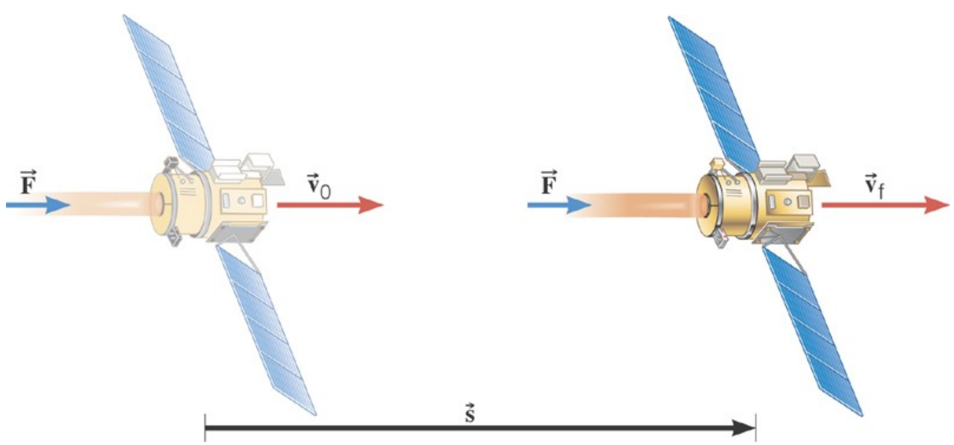
The mass of the space probe is 474-kg and its initial velocity is 275 m/s. If the 0.056 N force acts on the probe through a displacement of 2.42×10^{9}m, what is its final speed?



$$\left[\left(\sum \mathrm{F}\right)\cos\theta\right]s = \tfrac{1}{2}mv_{\mathrm{f}}^2 - \tfrac{1}{2}mv_o^2$$

$$\left(5.60\times10^{-2}\,\mathrm{N}\right)\cos 0^\circ\left(2.42\times10^{9}\,\mathrm{m}\right) = \tfrac{1}{2}\left(474\,\mathrm{kg}\right)v_{\mathrm{f}}^2 - \tfrac{1}{2}\left(474\,\mathrm{kg}\right)\left(275\,\mathrm{m/s}\right)^2$$

$$v_f = 805\,\mathrm{m/s}$$



1) In the 1950s, an experimental train that had a mass of 2.50 10^4 kg was powered
Across a level track by a jet engine that produced a thrust of 5 10^5 N for
A distance of 509m .
A) Find the work done on the train
B) Find the change in kinetic energy
C) Find the final speed of the train if there were no friction.(initial speed is 0)

2) A 2,000 kg has a speed of 12m/s. The car hits a tree. The tree does not move, and the car comes to rest.
A) Find the change in kinetic energy
B) Find the amount of work done in pushing in the front of the car
C) Find the size of the force that pushed into the front of the car by 50cm.

3) A constant lift of 410N is applied upward to a stone that weighs 32N. The Upward force is applied through 2m.
A) What is the net work done on the stone ?
B) What is the change in kinetic energy?
C) if the stone is 5 meters above the ground, what is its potential energy?

4) The force applied to the wagon is 100N @ 60
The wagon is pulled over a distance of 10m
A) what is the work done.
B) if there is a frictional force of 40N, what is the network?
C) change in KE

Try first without the hints ! Work is = change in energy = energy in transfer

7) You have a spring-loaded air rifle. When it is loaded, the spring is compressed 0.3m and has a spring constant of K =150N/m. In joules, how must elastic potential energy is stored in the spring? Hint: elastic PE= 0.5 K $(X)^2$ X is the distance compressed or stressed.

8) How much negative work in J is needed to slow a 588kg barge from a velocity of 1.1m/s to 0.4m/s? Hint: convert g to kg. Work is the change in KE.

9) An AR-15 rifle shoots a 4.10g bullet straight up with a velocity of 905m/s. It returns to the Earth a little while later with a velocity of 111m/s. How much negative work was done on the bullet by drag forces? Hint: work is the change in KE

10) How much work is required to compress a spring, k=33.3N/m, from its equilibrium point at x=0.0mto x=5.0m? (work = change in energy).

11) How much work is required to compress a spring (k=33.3N/m) from t x=5.0m to x=7.00m?

Conservation of energy

- The **Law of Conservation of Energy:** energy cannot be created or destroyed.
  - The total energy of an isolated system is constant.
    - The energy of the Universe is constant.
- Energy can only be transformed from one form to another.

Conservation of mechanical energy (PE and KE)

- If the energy of an isolated system is constant, the energy before an event must the same as the energy after an event.

$$\text{total energy before} = \text{total energy after}$$

Conservation of energy

- To deal with energy conservation, we need the total energy:

$$\text{total energy:}$$

$$E = KE + PE = \text{constant}$$

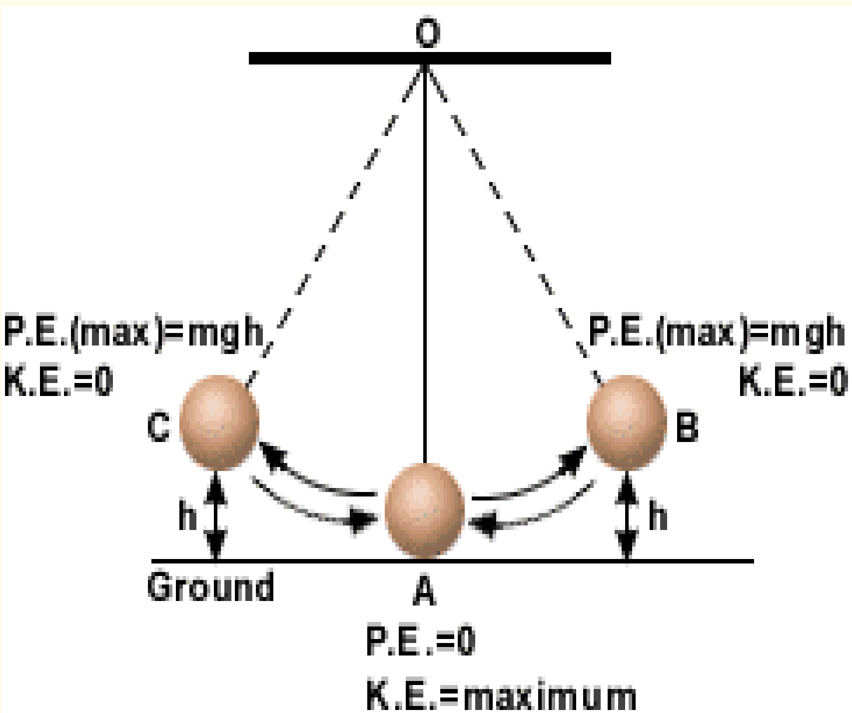
Simulations with exploration of physical sciences:

1) KE and PE of a falling ball
2) pendulum. Note that you need to decide of a level for which PE =0
3) spring and mass. Pel = 0.5 K x^2
4) frosty
5) jump

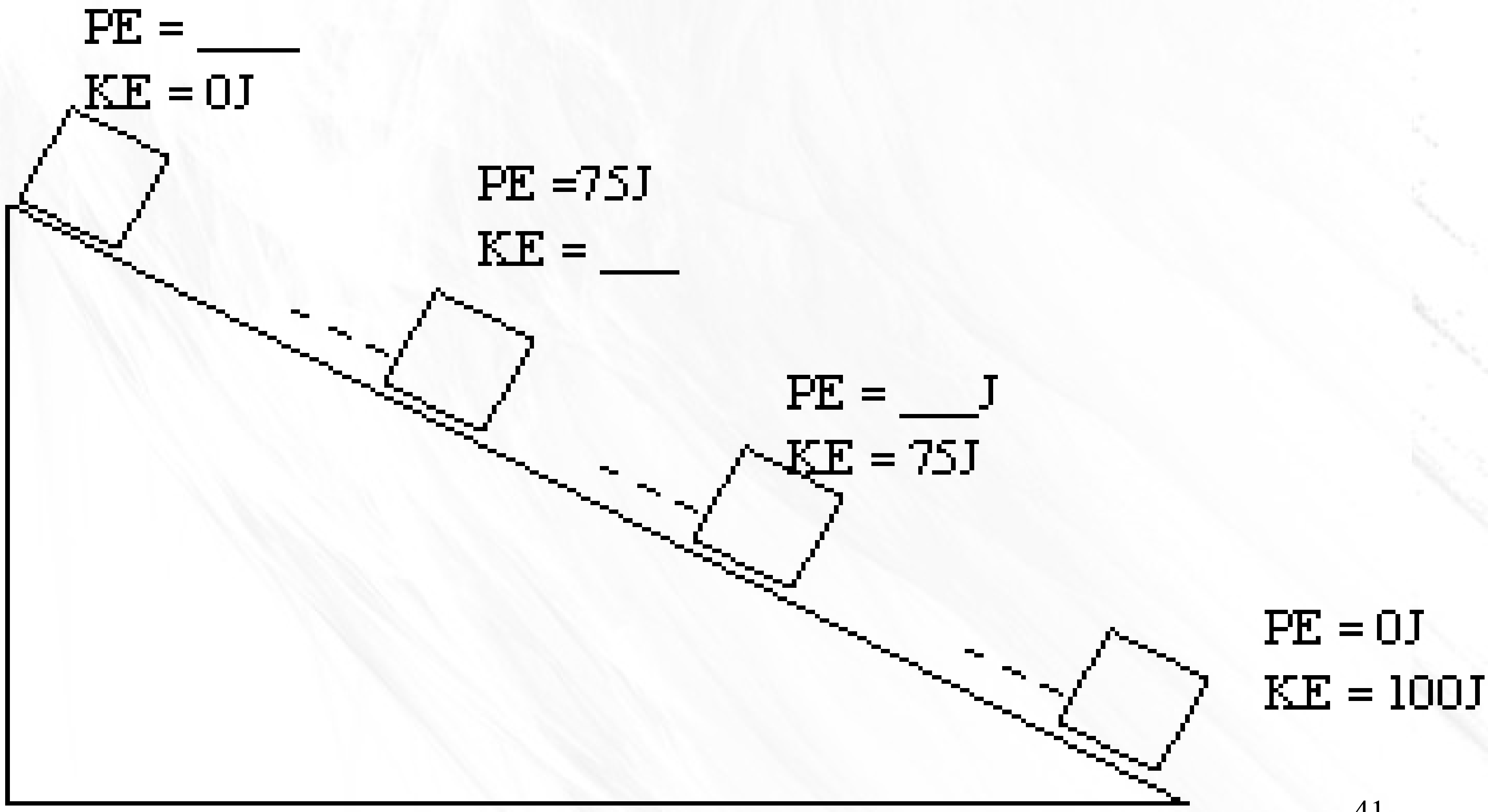
Source: http://www.physicscurriculum.com/

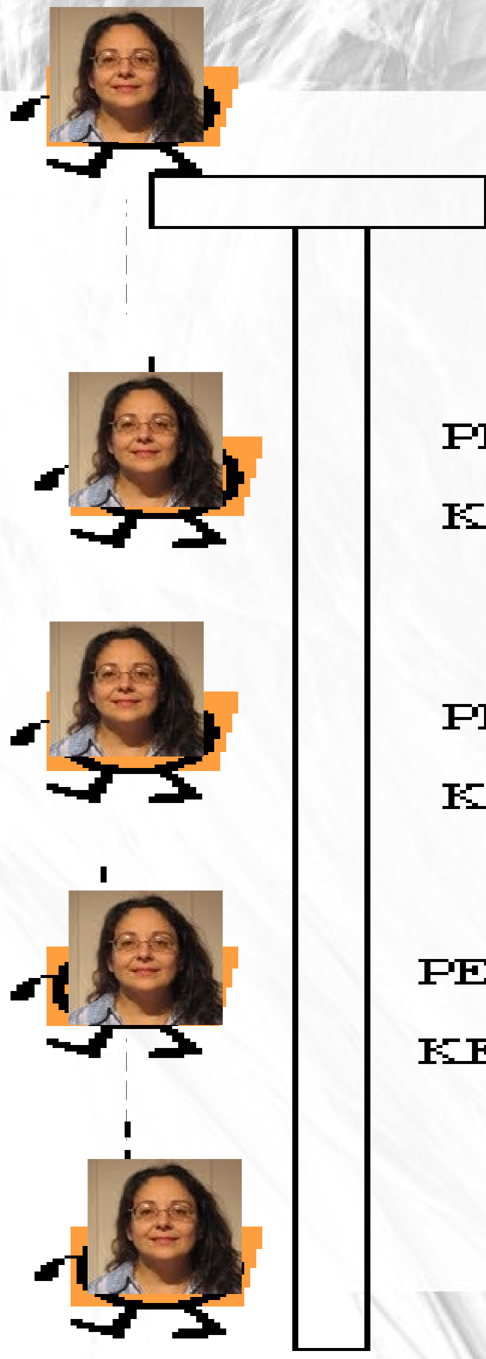
Conservation of energy

- We can understand a roller-coaster or a pendulum by as an example of energy conservation.



COMPLETE. A block slides on a frictionless plane:





PE = 15 000J

KE = 0

PE = 11,250J

KE = __________

PE = 7,500J

KE = _________

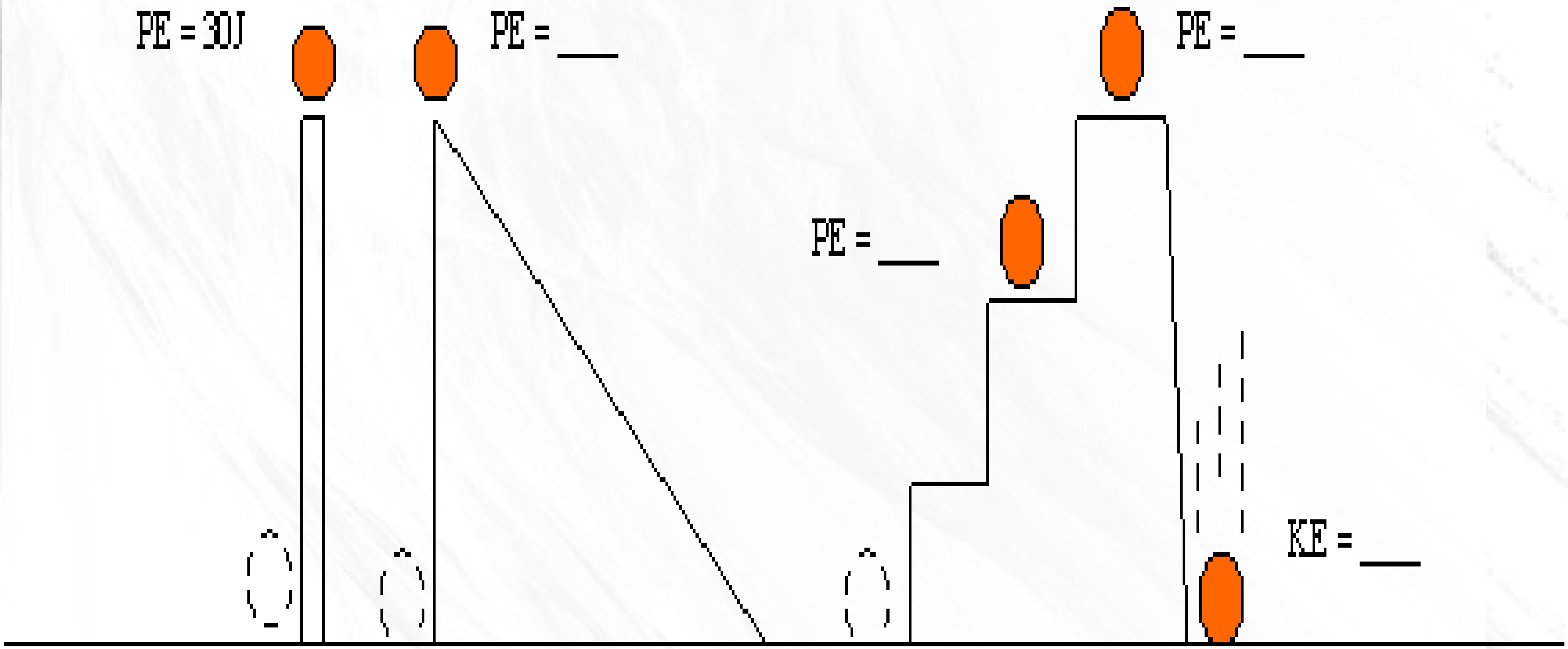
PE = 3,750 J

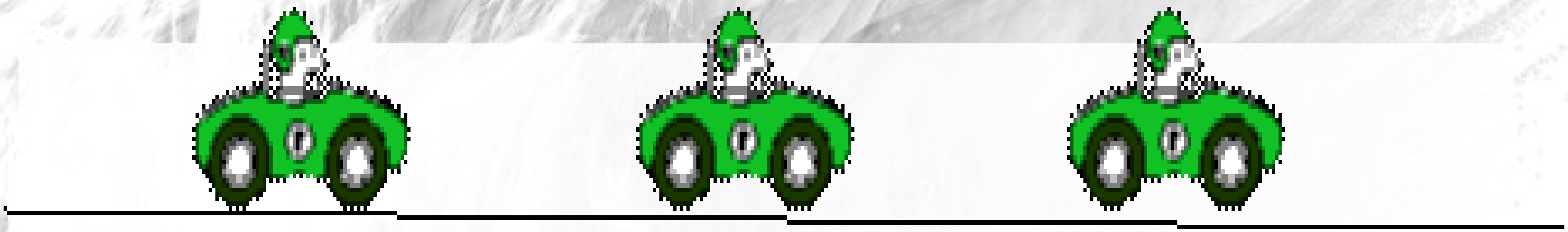
KE = _________

PE = 0 J

KE =

To get the ball at the top, you can roll it up the ramp or you can lift it. Will the change in energy be the same ? Why ?





V= 30km/h

KE = 10^6J

V= 60km/h

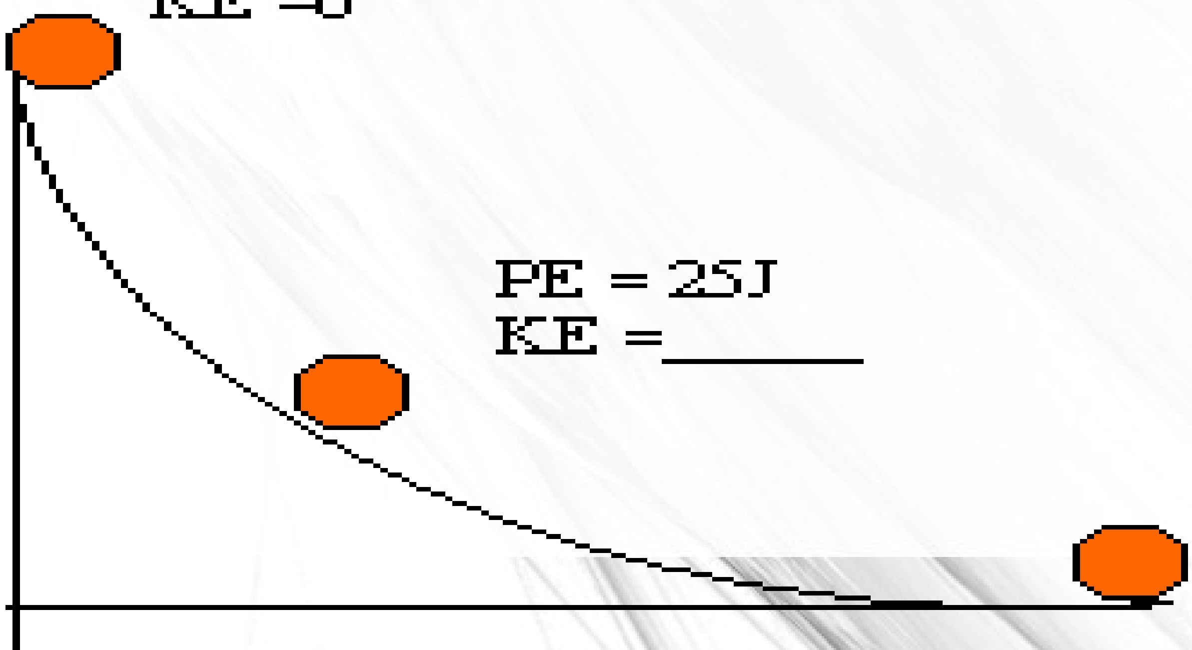
KE = ___

V= 90km/h

KE = ___

PE = ________

KE =0



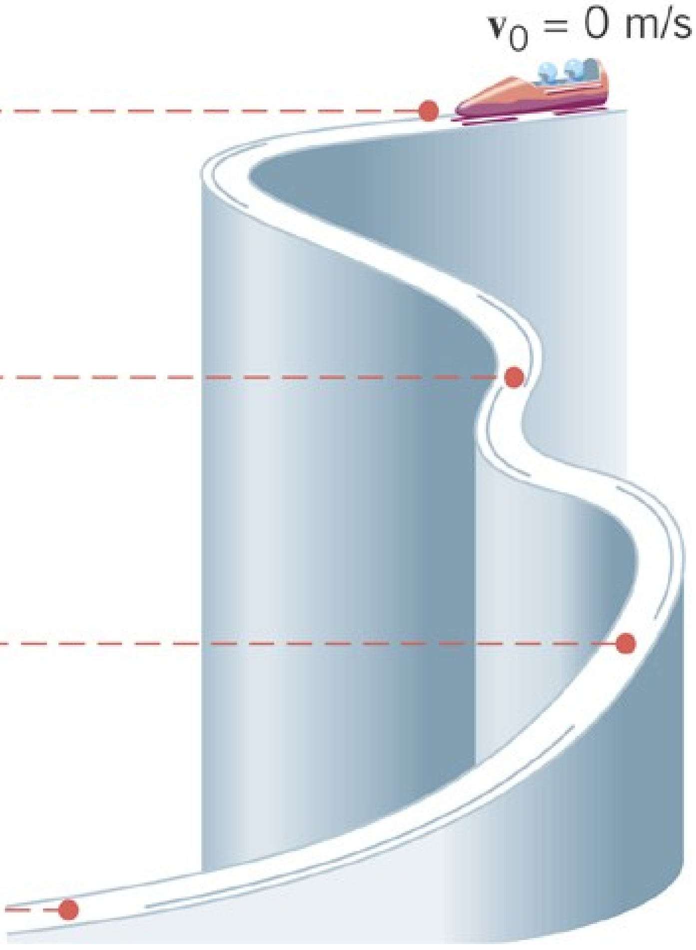
PE = 25J

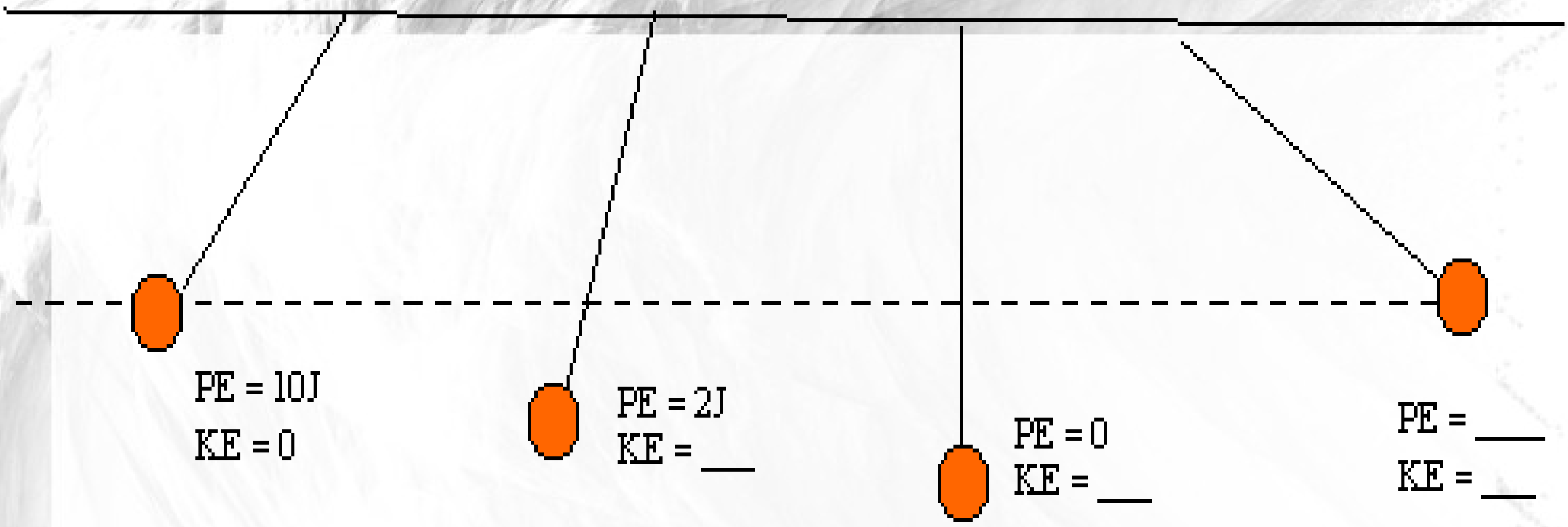
KE =______

PE = 0J

KE =50J

KE	PE	E = KE + PE
0 J	600 000 J	600 000 J
200 000 J	400 000 J	600 000 J
400 000 J	200 000 J	600 000 J
600 000 J	0 J	600 000 J





Starting from rest the cart glides friction less to point P, which is 4.5m below the top of the hill. How fast is it going at P ?

Videos in 101. conservation of mechanical energy

P

Example

In 2003, a man went over Horseshoe Falls, part of Niagara Falls, and survived. The height of the falls is about 50 meters. Estimate the speed of the man when he hit the water at the bottom of the falls.

In 2003, Jones pleaded guilty in Canada to a criminal charge of unlawfully performing a stunt and mischief and was fined $3,000.

http://www.youtube.com/watch?v=mhIOylZMg6Q

Video Wlater Lewin

Video what it goes wrong
App exploration of Physical sciences

DISCUSSION:

The speed does not depend on the man’s mass.

If you tried this, you’d hit the bottom with the same speed.

This is obviously ideal:

- We do not consider air resistance. That would convert some of his KE into heat and sound. The real speed would be slower.

Example

I toss a 0.06-kg tennis ball straight up. When it leaves my hand, it has a speed of 20 m/s. Find how high the ball rises.

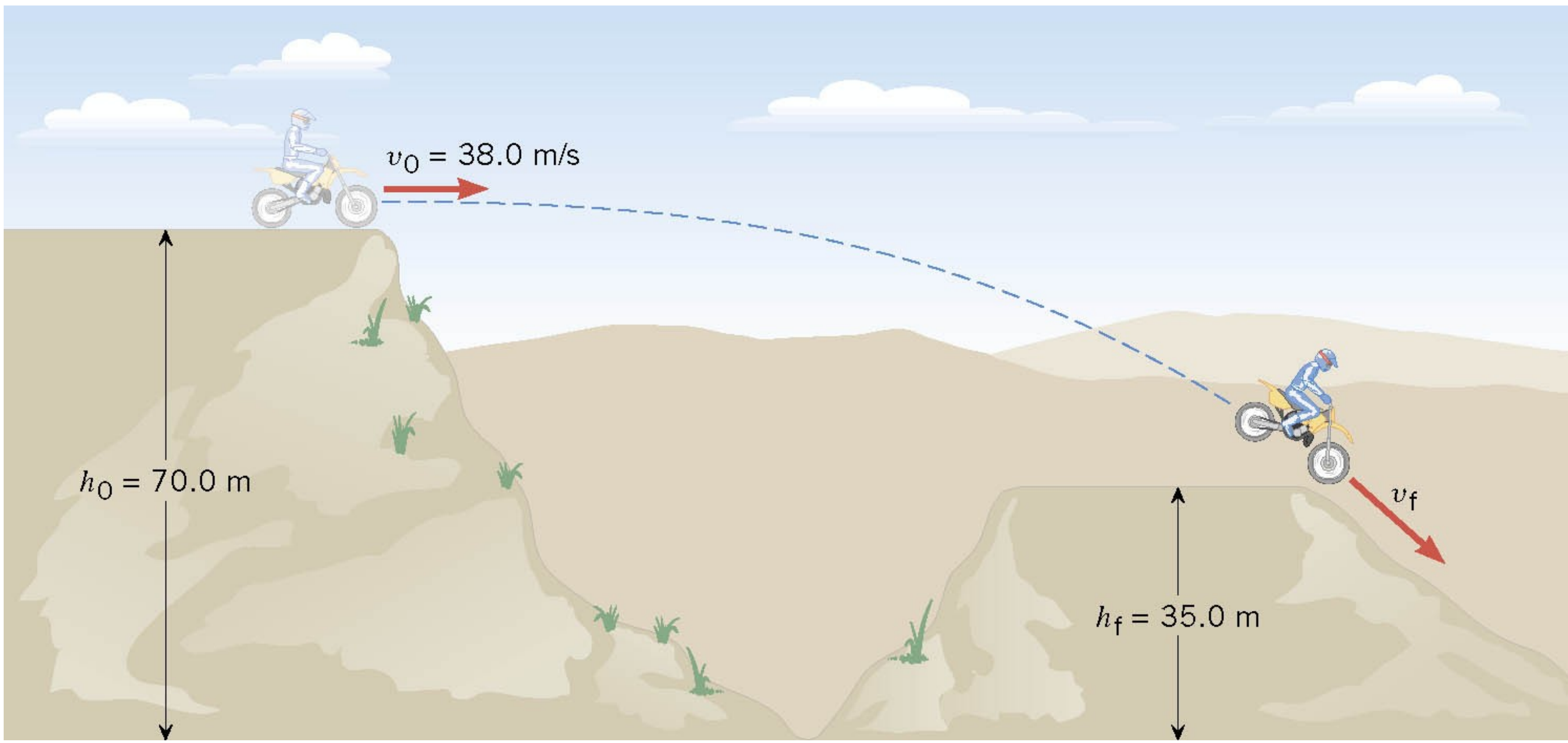
Note : you can use the kinematics equation for free-Fall

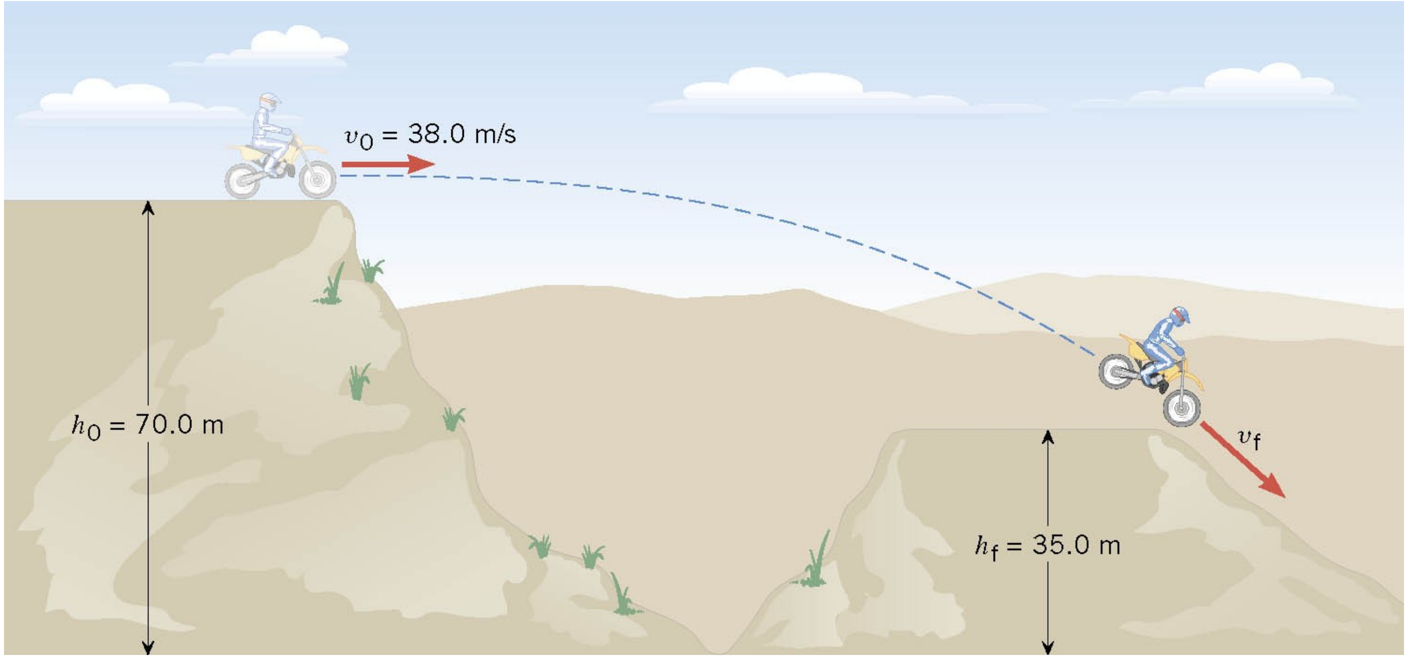
DISCUSSION:

Again, since we neglect air resistance the height would be the same if I tossed a tennis ball, bowling ball or brick.

Example 8 A Daredevil Motorcyclist

A motorcyclist is trying to leap across the canyon by driving horizontally off a cliff 38.0 m/s. Ignoring air resistance, find the speed with which the cycle strikes the ground on the other side.

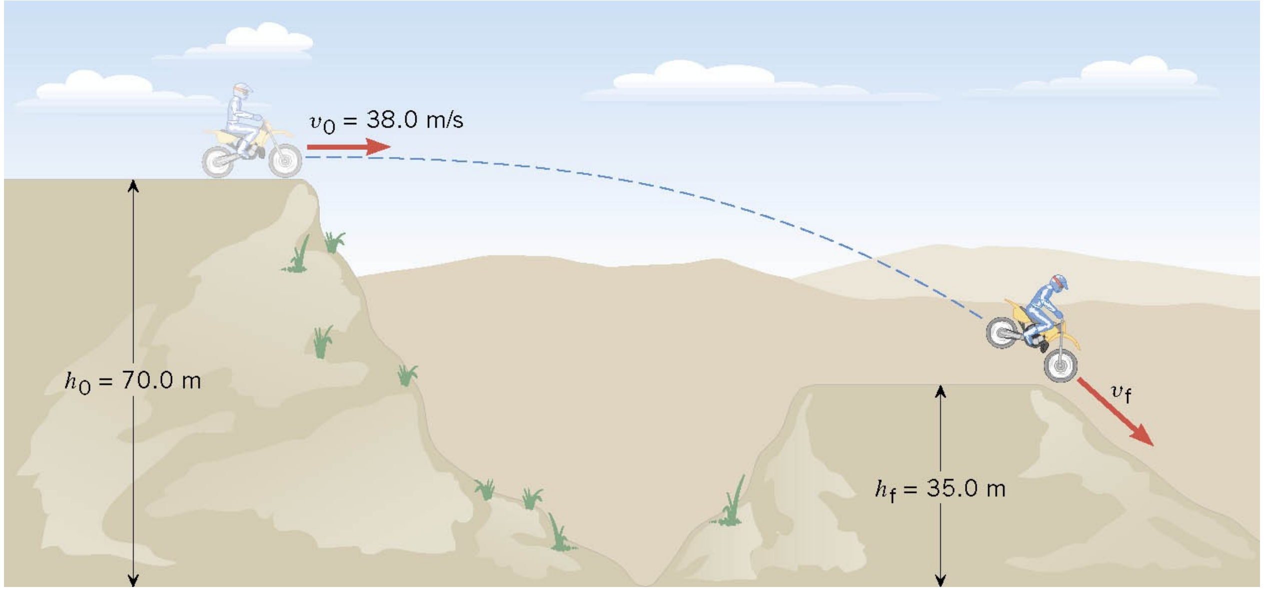




$$E_f = E_o$$

$$mgh_f + \tfrac{1}{2}mv_f^2 = mgh_o + \tfrac{1}{2}mv_o^2$$

$$gh_f + \tfrac{1}{2}v_f^2 = gh_o + \tfrac{1}{2}v_o^2$$



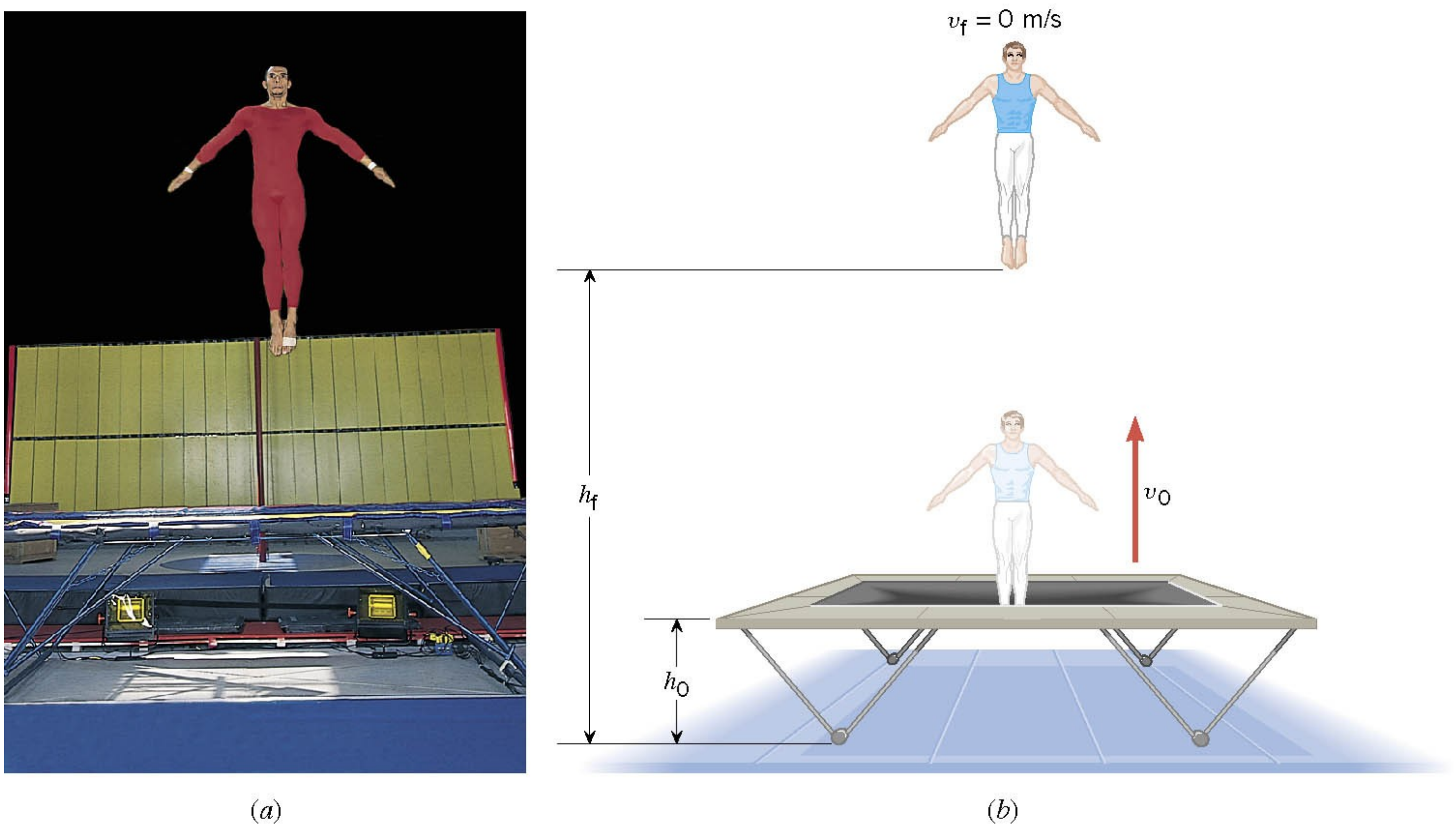
$$gh_f+\frac{1}{2}v_f^2=gh_o+\frac{1}{2}v_o^2$$

$$v_f=\sqrt{2g\left(h_o-h_f\right)+v_o^2}$$

$$v_f=\sqrt{2\left(9.8\,m/s^2\right)\left(35.0\,m\right)+\left(38.0\,m/s\right)^2}=46.2\,m/s$$

Example 7 A Gymnast on a Trampoline

The gymnast leaves the trampoline at an initial height of 1.20 m and reaches a maximum height of 4.80 m before falling back down. What was the initial speed of the gymnast?

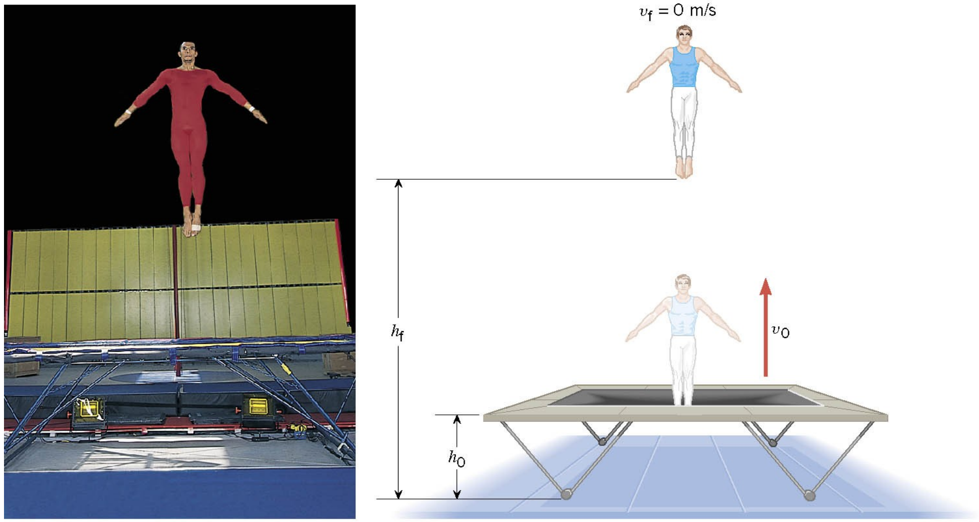


$$W = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_o^2$$

$$W_{\text{gravity}} = mg\left(h_o - h_f\right)$$

$$mg\left(h_o - h_f\right) = -\tfrac{1}{2}mv_o^2$$

$$v_o = \sqrt{-2g\left(h_o - h_f\right)}$$



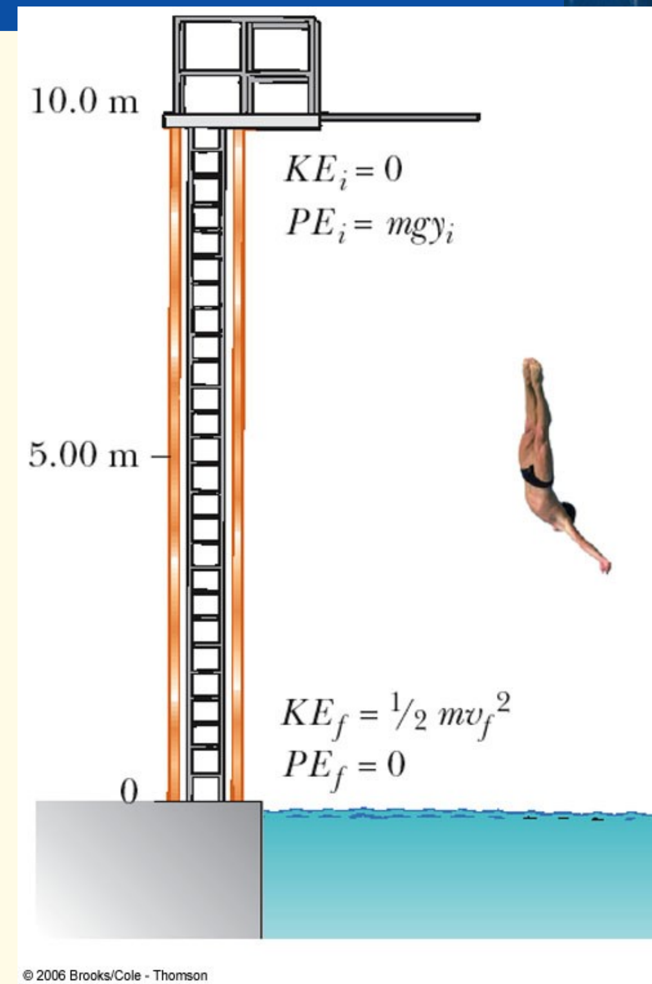
(a) (b)

$$v_o = \sqrt{-2\left(9.80\,\text{m/s}^2\right)\left(1.20\,\text{m} - 4.80\,\text{m}\right)} = 8.40\,\text{m/s}$$

Platform Diver

(a) Find his speed 5.0 m above the water surface

(b) Find his speed as he hits the water



Platform Diver

(a) Find his speed 5.0 m above the water surface

$$\frac{1}{2}mv_i^2 + mgy_i = \frac{1}{2}mv_f^2 + mgy_f$$

$$0 + gy_i = \frac{1}{2}v_f^2 + mgy_f$$

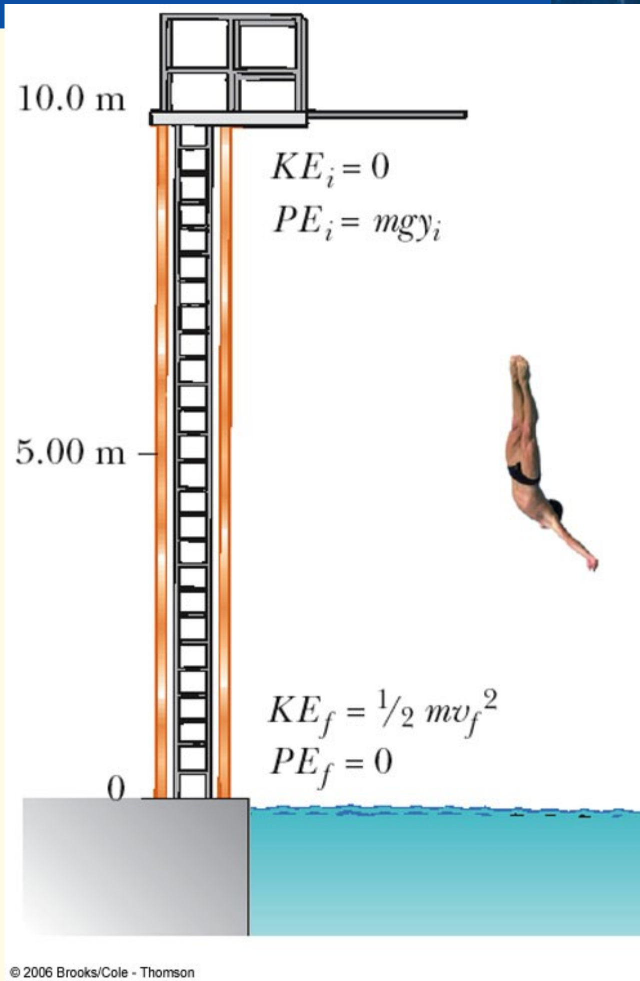
$$v_f = \sqrt{2g\left(y_i - y_f\right)}$$

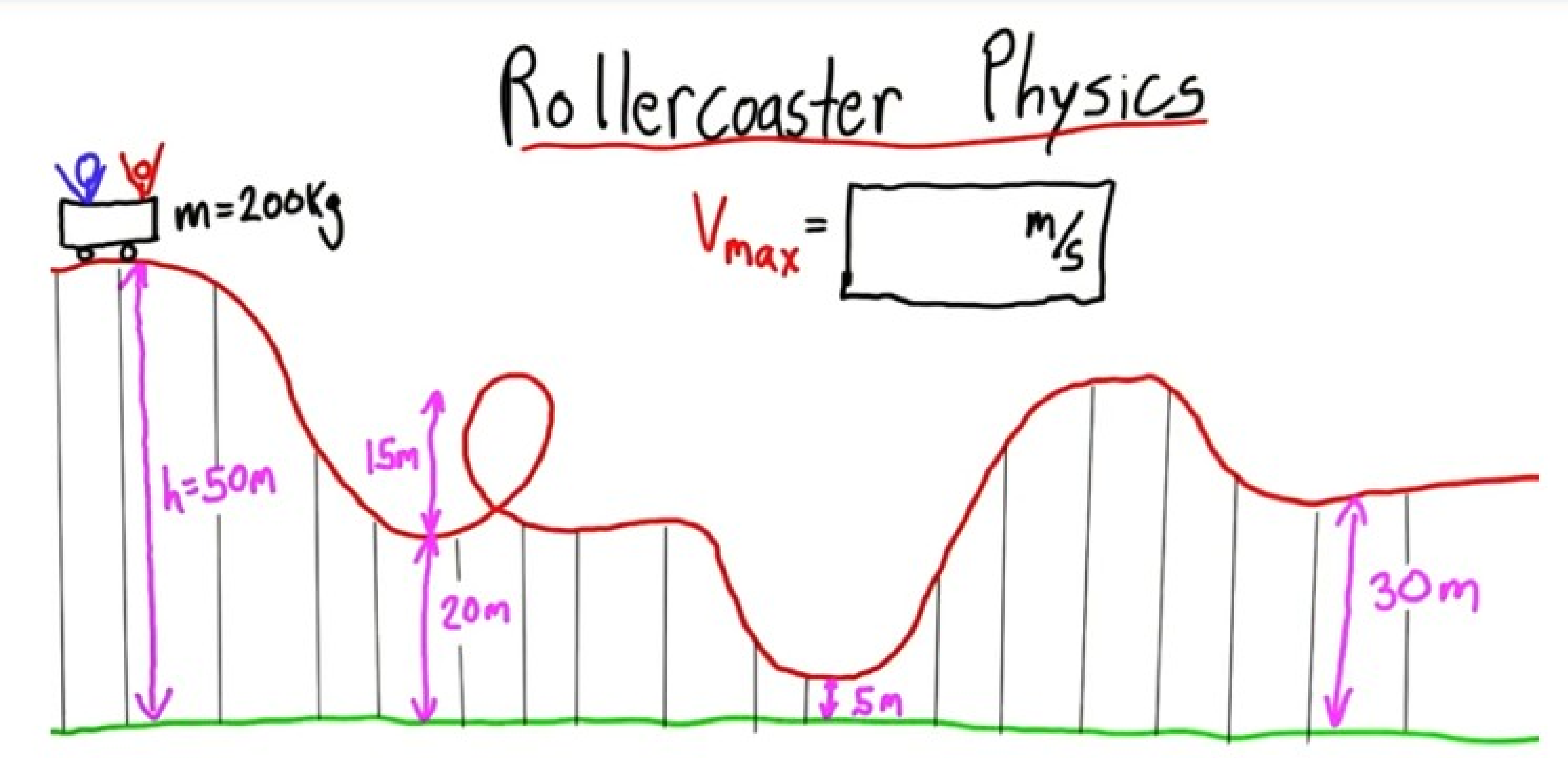
$$¿\sqrt{2\left(9.8\,m/s^2\right)\left(10\,m - 5\,m\right)} = 9.9\,m/s$$

(b) Find his speed as he hits the water

$$0 + mgy_i = \frac{1}{2}mv_f^2 + 0$$

$$v_f = \sqrt{2gy_i} = 14\,m/s$$




Rollercoaster Physics
m=200Kg
Vmax = m/s
h=50m
15m
20m
5m
30m

Non conservative Forces and the Work-Energy Theorem

If other forces than gravity are involved (thrust from engine or friction) then write:

PE1 + KE1 + (work done by thrust or you) = PE2 + KE2

Energy is added to the whole system. Ex If you push a cart

ex. : $mgh1 + 0.5\ m\ v1^2 + push\ x\ displacement = mgh2 + 0.5\ m\ V2^2$

This is the same as : net work = change in KE

PE1 + KE1 + (work done by friction) = PE2 + KE2

Work done by friction is negative = - friction x displacement

Energy is removed from the whole system. Ex: friction

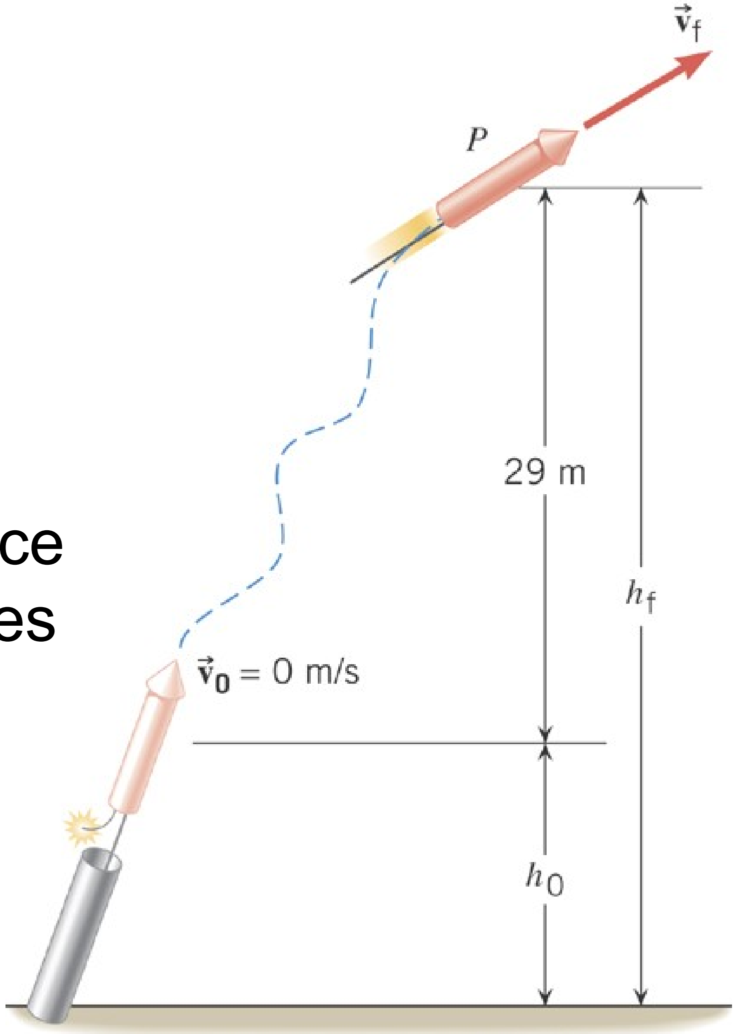
$mgh1 + 0.5\ m\ v1^2 - friction\ x\ displacement = mgh2 + 0.5\ m\ V2^2$

Friction= μk Normal.μk is the coefficient of kinetic friction

Can be work done by air resistance. The work is also negative.

Example 11 Fireworks

Assuming that the nonconservative force generated by the burning propellant does 425 J of work, what is the final speed of the rocket. Ignore air resistance.



6.6 Nonconservative Forces and the Work-Energy Theorem

$$W_{nc} = mgh_f - mgh_o + \tfrac{1}{2}mv_f^2 - \tfrac{1}{2}mv_o^2$$

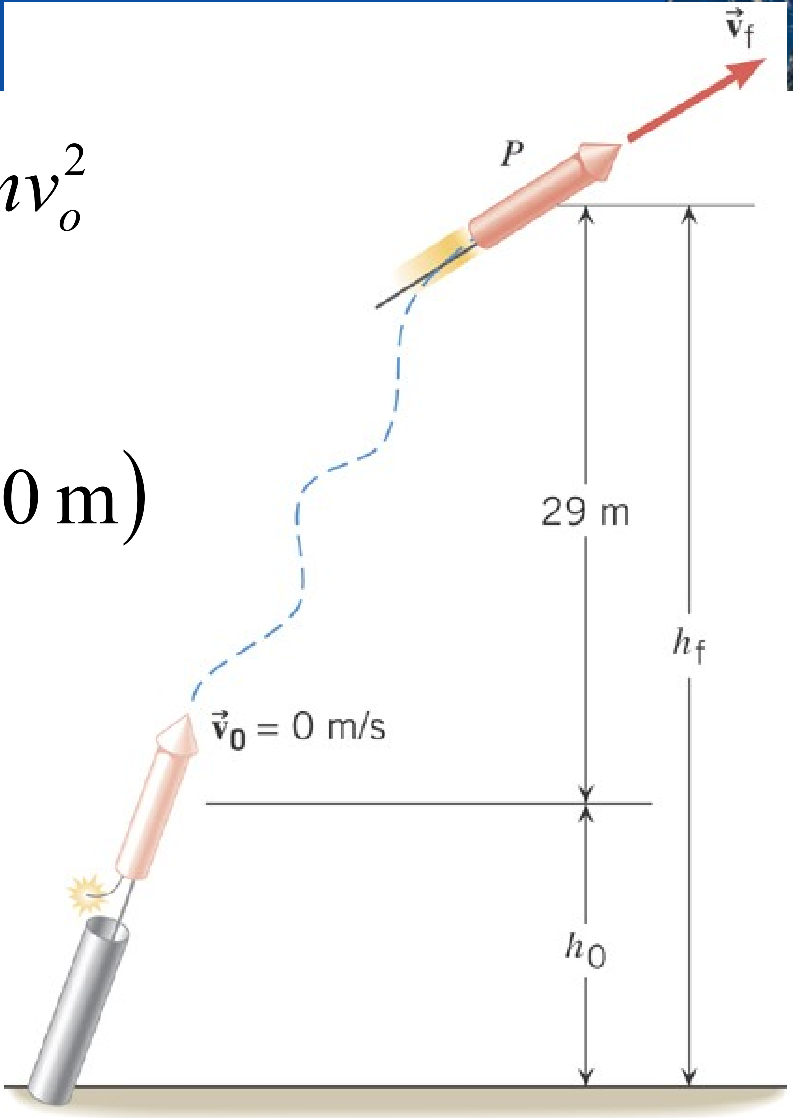
$$W_{nc} = mg\left(h_f - h_o\right) + \tfrac{1}{2}mv_f^2$$

$$425\,\text{J} = (0.20\,\text{kg})\left(9.80\,\text{m/s}^2\right)(29.0\,\text{m})$$

$$+\tfrac{1}{2}(0.20\,\text{kg})v_f^2$$

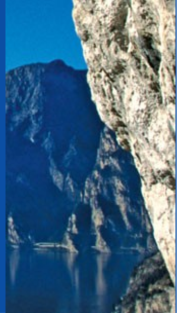
$$v_f = 61\,\text{m/s}$$

Show picture pumping half life



Nonconservative Forces and the Work-Energy Theorem

PE1 + KE1 - (work done by friction) = PE2 + KE2



An example of a nonconservative force is the kinetic frictional force.

$$W = \left(F\cos\theta\right)s = f_k \cos 180^{\circ} s = -f_k s$$

The work done by the kinetic frictional force is always negative. Thus, it is impossible for the work it does on an object that moves around a closed path to be zero.

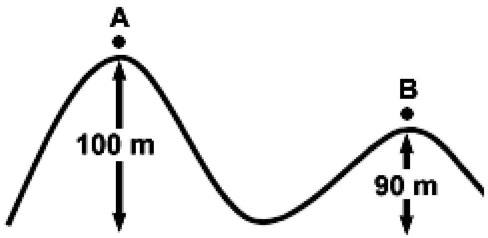
1) Which has the greater kinetic energy - a 1-ton car moving at 30 m/s, a half-ton car moving at 60 m/s?

2) A 20 N ball and a 40 N ball are dropped at the same time from a height of 10 meters.
Air resistance is negligible.
A)Compute the speed of impact for each object.
B)change of momentum of each object and impulse at impact

3) Jim exerts a force of 500 N against a 100-kg desk that does not move.
Virgil exerts a force of 400 N against a 60-kg desk that moves 2 m in the direction of the push.
Mik exerts a force of 200 N against a 50-kg desk that moves 4 m in the direction of the push.
The most work is done by ?

4) Suppose you climb the stairs of a ten-story building, about 30 m high, and your mass is 60 kg. The gravitational potential energy you gain is about

5) A skier starts from rest at A. What is his speed at B ?



http://www.sparknotes.com/physics/workenergypower/conservationofenergy/problems_1.html

ESCAPE VELOCITY

Fire bullets up and they will come back down ! (some people don't realize that).

You need 7 miles per second to escape Earth gravity. (11km/s).
Interestingly, an object from infinity falls on earth with the same speed.

To find the speed : gravitation energy = kinetic energy.
Solve for the speed.

$$\frac{1}{2}mv_i^2 - \frac{GM_E m}{R_E} = \frac{1}{2}m0^2 - \frac{GM_E m}{\infty}$$

$$\frac{1}{2}mv_i^2 - \frac{GM_E m}{R_E} = 0 - 0$$

$$\frac{1}{2}mv_i^2 - \frac{GM_E m}{R_E} = 0$$

$$V_e = \sqrt{\frac{2GM}{R}}$$

$$= \sqrt{\frac{2 \times 6.6726 \times 10^{-11} \times 5.9742 \times 10^{24}}{6378100}}$$

$$= \sqrt{\frac{2 \times 6.6726 \times 5.9742 \times 10^{24-11}}{6378100}}$$

$$= \sqrt{\frac{79.73 \times 10^{13}}{6378100}}$$

$$= \sqrt{125005879.5} = 11180.6 \text{ m/s}$$

http://highered.mcgraw-hill.com/olcweb/cgi/pluginpop.cgi?it=swf::800::600::/sites/dl/free/0072482621/78778/Escape_Nav.swf::Escape%20Velocity%20Interactive

Table 1.1 Energy per Gram

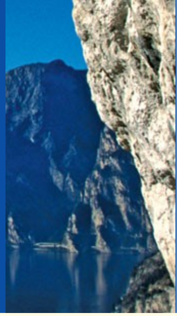
Object	Calories (or watt-hours)	Joules	Compared to TNT
Bullet (at sound speed, 1000 ft/s)	0.01	40	0.015
Battery (auto)	0.03	125	0.05 – x 2
Battery (rechargeable computer)	0.1	400	0.15 – x 1
Flywheel (at 1 km/s)	0.125	500	0.2
Battery (alkaline flashlight)	0.15	600	0.23
TNT (the explosive trinitrotoluene)	0.65	2700	1
Modern high explosive (PETN)	1	4200	1.6
Chocolate chip cookies	5	21,000	8
Coal	6	27,000	10
Butter	7	29,000	11
Alcohol (ethanol)	6	27,000	10
Gasoline	10	42,000	15
Natural gas (methane, CH_4)	13	54,000	20
Hydrogen gas or liquid (H_2)	26	110,000	40
Asteroid or meteor (30 km/s)	100	450,000	165

DEFINITION OF AVERAGE POWER

Average power is the rate at which work is done, and it is obtained by dividing the work by the time required to perform the work.

$$\overline{P} = \frac{\text{Work}}{\text{Time}} = \frac{W}{t}$$

$$\text{joule/s} = \text{watt (W)}$$



Power

Power is the rate at which energy is transferred or the rate at which work is done.

$$P = \frac{W}{t}$$

$$P = \frac{\Delta E}{t}$$

P = power (Watt)
W = work done (J)
ΔE = energy transferred (J)
t = time (s)

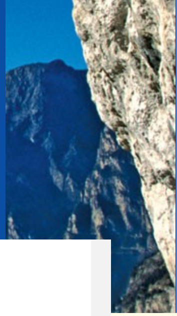
When a car stops, 40000J of work is done by the brakes in a time of 5s. Calculate the power of the brakes.

Table 6.4 Human Metabolic Rates[a]

Activity	Rate (watts)
Running (15 km/h)	1340 W
Skiing	1050 W
Biking	530 W
Walking (5 km/h)	280 W
Sleeping	77 W

[a]For a young 70-kg male.

We are like a 100W light bulb !!
So we spend 100 J/ second.
Compute how much energy do we spend in 1 day.
Convert to food calories. 4,184 joules = 1 Cal



for power examples

1 watt for flashlight battery

100 watts for a human or incandescent light

1 hp for a typical horse or athlete exercising for short period of time

1 kilowatt for a small house = **energy from the Sun for a square 1 m x 1m**

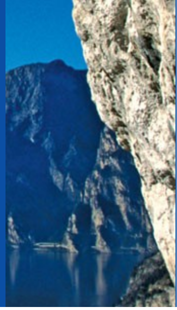
100hp for a small car

1 megawatt for electric power for small town (1000 houses)

1 gigawatt for large power plant (1000,000)

2 terawatts for the world (4/10 for the US)

Source: Richard A. Muller, Physics for future presidents.



Each year 500 runners run up the stairs to the 86th floor of the Empire State Building in New York Cit
There are 1576 steps and each step is 0.28 m high. In 2003,
Australian Paul Crake (20-29 age group) set the overall record by reaching the 86th floor in 9:33.
His mass was 70.0 kg.

Find the energy used by the racer. And the power

Kwh is a unit of energy :

Multiply the power (in kW) by the time (in hours)

Example: a microwave use 800 watts. If you run it for 2 hours and if the electric company charges 10 cents/kwh. How much you are going to pay ?

Hint energy(kWh) = power (kW) x time(hours)

P = 0.8 kW time = 2hours so E = 1.6kWh
so $$ = 16 cents.

Electricity is cheap !

A large household air conditioner may consume 15.0kW of power. What is the cost in dollars of operating this air conditioner 3.00h per day for 30.0d if the cost of electricity is $0.110 per kWh?

In Miami:

A large central air conditioning system for a home has a power rating of 5000 watts. Assuming it runs for an average of 10 hours per day for 30 days, and the electricity cost is $0.14 per kWh, what is the monthly cost of running this AC system?

A small refrigerator has a power rating of 100 watts. Assuming it runs 24 hours a day, 30 days a month, and the cost of electricity in Miami is $0.14 per kilowatt-hour (kWh), what is the monthly cost of running this refrigerator?

Other unit for power is horsepower or hp

Introduced by James Watt.

A horse can lift 330 pounds weight vertically

For a distance of 100 ft in one minute

Or P = 330 x 100 foot-pounds per minute

Or P = 33,000 foot-pounds per minute

We can produce the same power but for a short time.

Usually, we can produce 4/7 hp with sustained effort.

(like heavy duty exercise not yoga)

1/7 is used to do work the other part is lost to heat.

(1/7 is 25% of 4/7, so we are only 25% efficient)

1Hp is 746 W about 1kW

See p. 20 table

So 1 horse can lift ___________ pounds over 1 foot

So we can produce 4/7 hp during a sustained exercise.

BUT NOT FOR VERY LONG !!
Unlike a horse.

New definition for Average Power delivered by a constant force (a constant acceleration). But not constant speed :

$$\text{Power} = \frac{\text{Work}}{\text{Time}} = \frac{\text{Force} \bullet \text{Displacement}}{\text{Time}}$$

$$\text{Power} = \text{Force} \bullet \frac{\text{Displacement}}{\text{Time}}$$

$$\boxed{\text{Power} = \text{Force} \bullet \text{Velocity}}$$

Power Delivered by an Elevator Motor

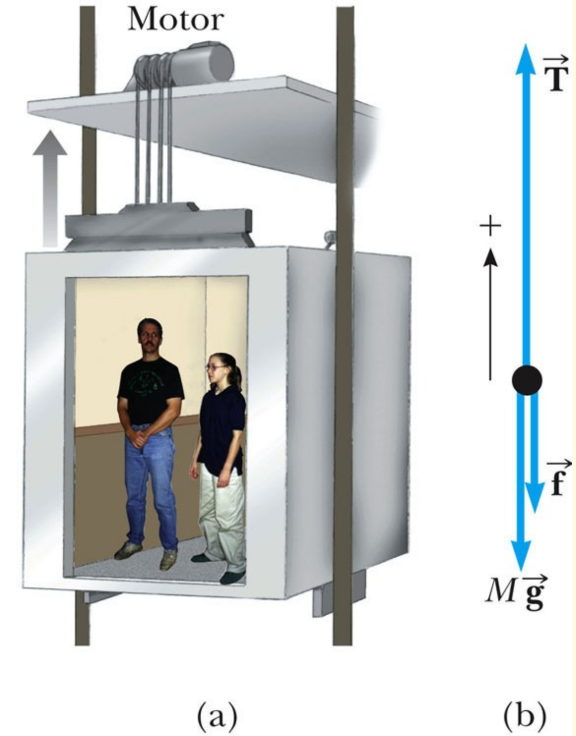
- A 1000-kg elevator carries a maximum load of 800 kg. A constant frictional force of 4000 N retards its motion upward. The elevator moves at a constant speed of 3 m/s?

  1) What is the force T lifting the elevator? (up = down)

  2) Use the equation

  average power = force x average speed

  To find the minimum power delivered by T. (delivered to lift the fully loaded elevator at a speed of 3m/s)



Power Delivered by an Elevator Motor

- A 1000-kg elevator carries a maximum load of 800 kg. A constant frictional force of 4000 N retards its motion upward. What minimum power must the motor deliver to lift the fully loaded elevator at a constant speed of 3 m/s?

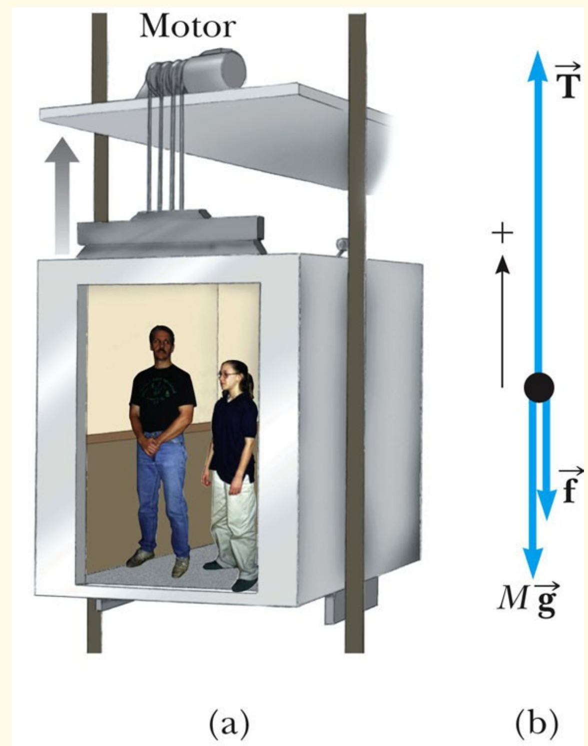
$$T - f - Mg = 0$$

$$T = f + Mg = 2.16 \times 10^4\ N$$

$$P = Fv = (2.16 \times 10^4\ N)(3\ m/s)$$

¿ $6.48 \times 10^4\ W$

$$P = 64.8\ kW = 86.9\ hp$$



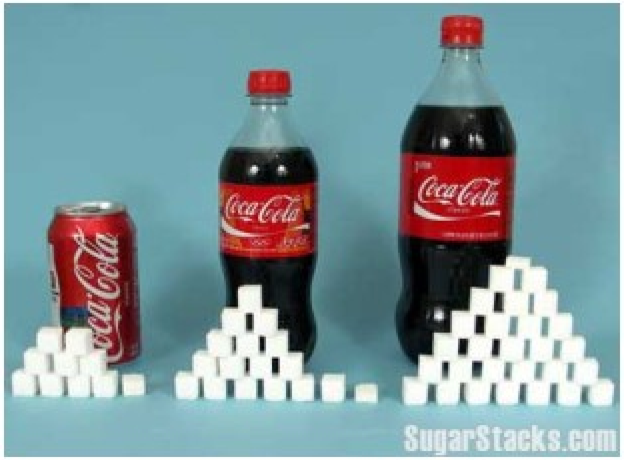
Units for energy

1 Kwh is a unit for energy = 1000 x 3600 J
or about 4 million J
1 good calorie or 1 Cal = 4000J so 1 Cal is 1Wh about
Or 1 KWh is about 4000 Cal.
Don't confuse the chem cal and the food Cal. 1 Cal = 1000 cal

We are like a 100W light bulb.
In one day we use about ______ Wh
Which is about ___________ Cal (1Wh is 1 Cal).

Lots of energy in food! 1 glass of milk = 150Cal = 1 soda can.
It takes 30 min hard gym to burn 150 Cal.

Can you lose weight doing exercises ?
Or is it better to cut on Food ? What's your guess ?



1 can of soda is about 150 Calories
(so is 1 glass of orange juice or milk)

You can burn this much of calories
In half an hour of cardio exercise

In your body you have 7 Cal per gram.
So if you cut your Calories from 2,000 to 1,500 calories per day.
How many grams do you lose per day ?
Per week ? How many pounds per week ? (500g = 1 pound)

If you exercise for 1 hour how many calories do you lose ?
How many grams is that ? (compare with previous)
So in a week you lose ? So?

Reference: Physics for future presidents – Richard Muller

Energy and Power Units

Energy units

- joules (J)
- calories (cal, kcal)
- kilowatt-hours (kWh)
- British thermal units (Btu)
- electron-volts (eV)
- Barrels of oil equivalent (BOE)

Power units

- watts (W; 1 W = 1 J/s)
- horsepower (hp; 1 hp=746 W)
- Million barrels of oil equivalent per day (MBOE/day)
- Humankind's energy consumption rate: 16 TW (16×10^{12} W)